Reference Schematics For RK3588S

RK3588S_Tablet_Demo_SCH

Main Functions Introduction

1) Charger: 1Cell Battery_QC
2) PMIC: 1 x RK806-1+DiscretePower
3) RAM: 2 x 32bits LPDDR4/4x
4) ROM: eMMC5.1(Default)
5) Support: 1 x Type-C 3.0(with DP function)
6) Support: 1 x 4Lanes MIPI D/CPHY RX Camera
7) Support: 1 x 2Lanes MIPI DPHY RX Camera
8) Support: 1 x 4Lanes MIPI D/CPHY TX
9) Support: a/b/g/n/ac/ax 2T2R WIFI(PCIE) + BT5.0
11) Support: 1 x Headphone + 2 x Speaker out + 1 x Analog MIC
12) Support: 2 x PDM MIC Array
13) Support: Gyroscope+G-sensor+Ambient Light+Proximity

MURAT IRKAN 2024/02

Table of Content

Page 1	00.Cover Page
Page 2	01.Index and Notes
Page 3	02.Revision History
Page 4	03.Block Diagram_PoweTree-1cell
Page 5	05.System Power Sequence
Page 6	06.Lower-Speed Bus Map
Page 7	07.USB Controller Configure Tab
Page 8	08.PCIE Fun Map
Page 9	10.RK3588S_Power/GND
Page 10	11.RK3588S_OSC/PLL/PMUIO
Page 11	12.RK3588S DDR Controler
Page 12	13.RK3588S Flash/SD Controller
Page 13	14.RK3588S_USB20/USB30/DP PHY
Page 14	15.RK3588S_SARADC/1.8V GPIO
Page 15	16.RK3588S_MIPI Interface
Page 16	17.RK3588S_HDMI/eDP Interface
Page 17	18.RK3588S PCIE2/SATA3/USB3 PHY
Page 18	19.RK3588S GPIO
Page 19	20.Power_1Cell_QC
Page 20	22.Power-PMIC_RK806-1
Page 21	23.Power_Ext Discrete
Page 22	24.RTC
Page 23	38.DRAM-LPDDR4/4X_200P_2X32bit
Page 24	40.eMMC Flash
Page 25	46.VI-Camera MIPI CSI0-RX
Page 26	47.VI-Camera_MIPI-DPHY-RX
Page 27	55.VO-MIPI DPHY-TX
Page 28	63.WIFI/BT-PCIe_2T2R(AP6275PR3)
Page 29	70.Audio Codec-ES8388
Page 30	90.Sensor
Page 31	92.KEY Array
Page 32	93.Debug UART/JTAG Port
Page 33	
Page 34	
Page 35	
Page 36	
Page 37	
Page 38	
Page 39	
Page 40	
Page 41	
Page 42	
Page 43	
Page 44	
Page 45	
Page 46	
Page 47	
Page 48	
Page 49	
Page 50	
Page 51	
Page 52	

Note

The power suffix S0 or S3 means:

S3: Keep power On during sleeping

S0:Power off during sleeping

Generate Bill of Materials

Header:

Item\tPart\tDescription\tPCB Footprint\tReference\tQuantity\tOption

Combined property string:

{Item}\t{Value}\t{Description}\t{PCB Footprint}\t{Reference}\t{Quantity}\t{Option}

Description

Note

Option

Notes

NOTE 1:
Component parameter description
1. DNP stands for component not mounted temporarily
2. If Value or option is DNP, which means the area is reserved without being mounted

NOTE 2:
Please use our recommended components to avoid too many changes.
For more informations about the second source,please refer to our AVL.

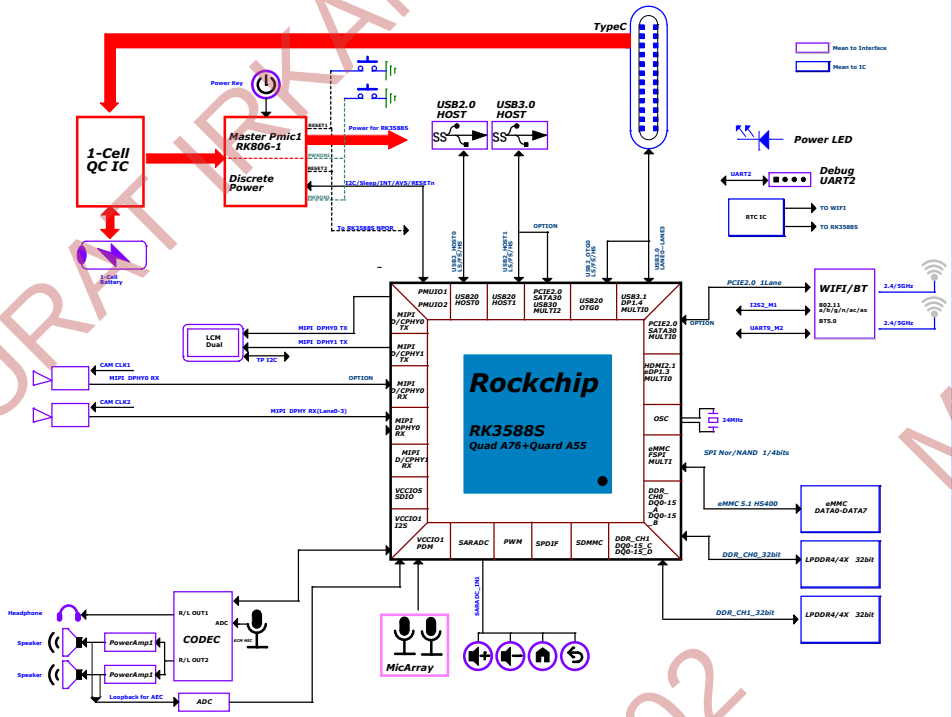
MURAT IRKAN 2024/02

Revision History

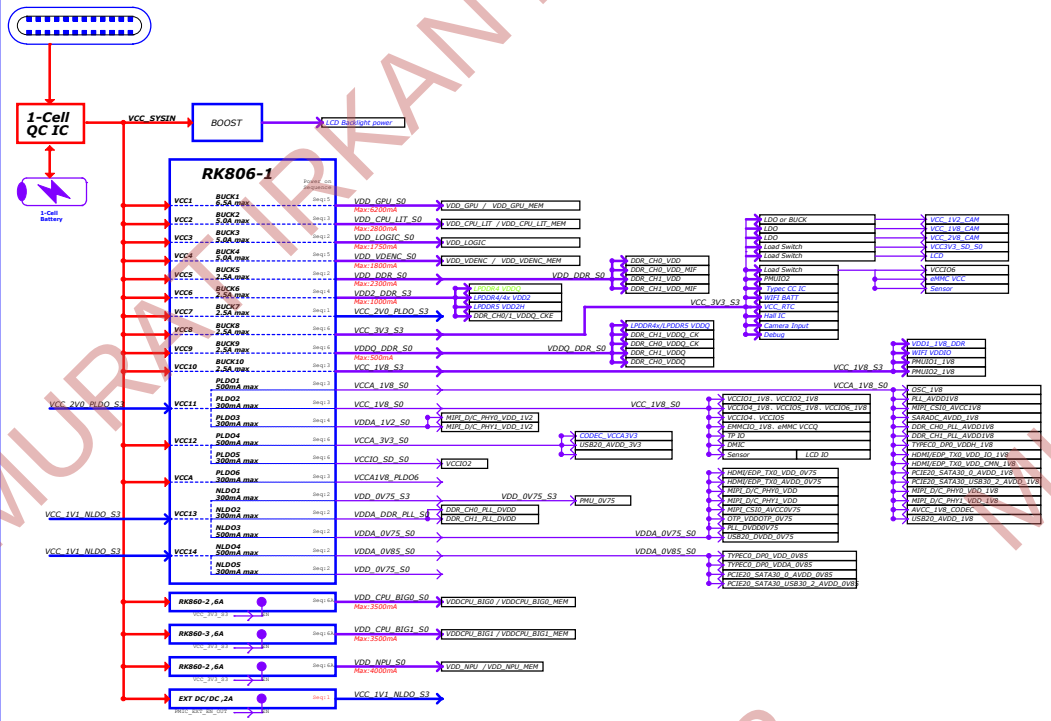
Version	Date	By	Change Dsecription	Approved
V1.0	2022-01-05	Joseph.Wei	1:Revision preliminary version	
V1.1	2022-02-18	Joseph.Wei	1.C1604,C1612的电容改成1uF/4V。 2.为了减少待机功耗，将PMUIO2电源域改成1.8V，此IO域对应外设IO电压相应修改 3.把L2203，L2205，L2207，L2300，L2301，L2302电感由0.22uH(TDK)改为0.24uH(Sunlord); L2201的电感由0.22uH(TDK)改为0.22uH (Sunlord)，封装IND_404020。	

MURAT IRKAN 2024/02

RK3588S Tablet Demo Block Diagram for 1-Cell Charger

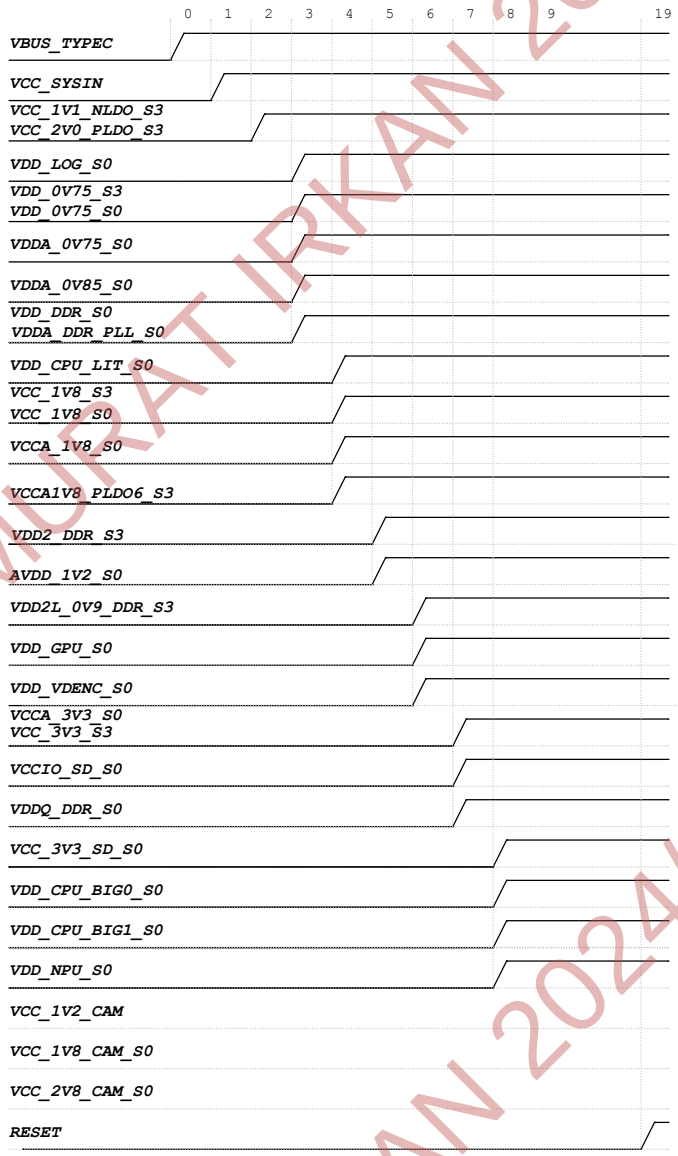


Power tree for 1-Cell Charger



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Power Sequence



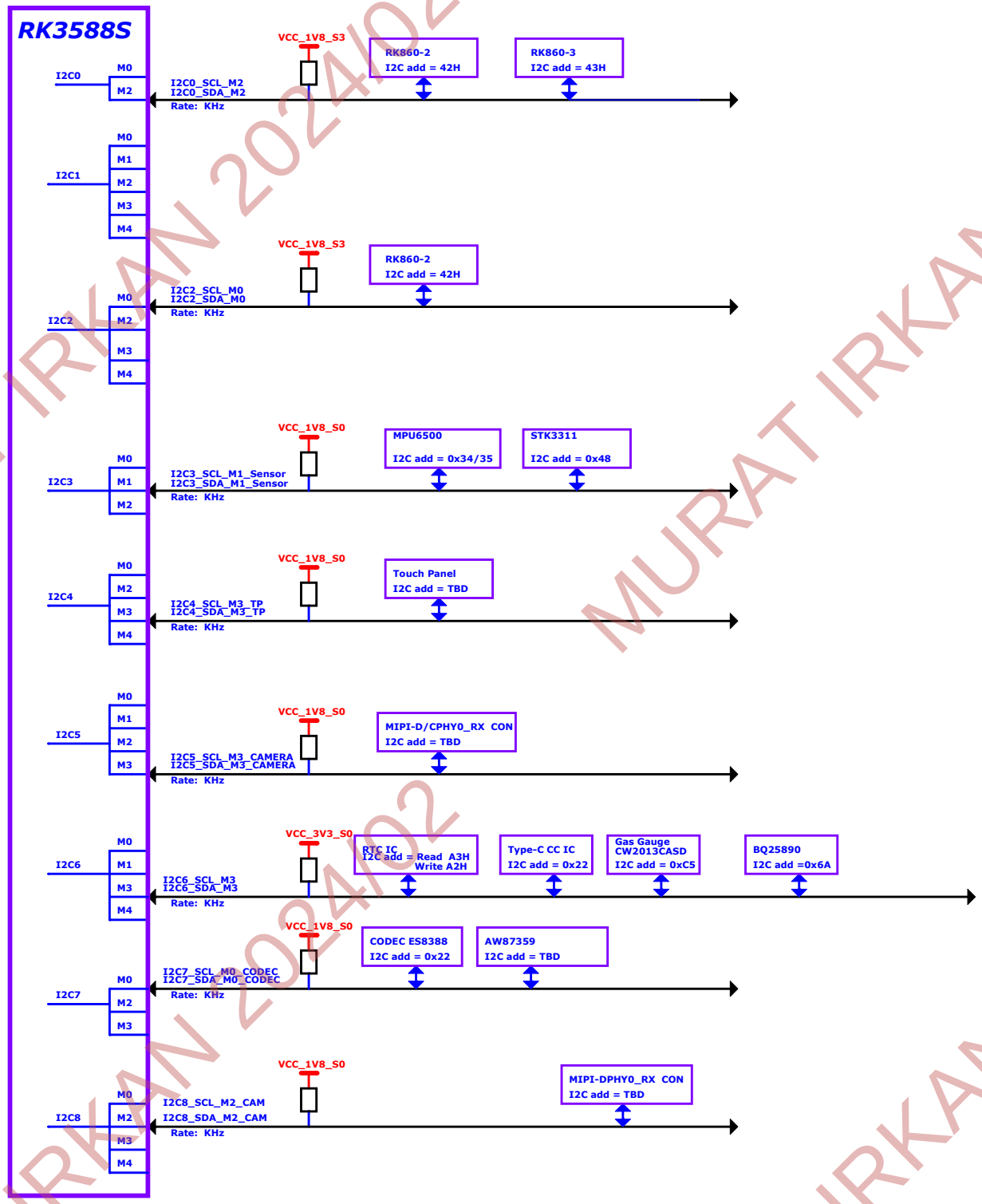
Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Sleep ON/OFF	Peak Current	Sleep Current
VCC_SYSIN	RK806-1_BUCK1	6.5A	VDD_GPU_S0	Slot:5	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK2	5A	VDD_CPU_LIT_S0	Slot:3	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK3	5A	VDD_LOG_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK4	3A	VDD_VDENC_S0	Slot:5	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK5	2.5A	VDD_DDR_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK6	2.5A	VDD2_DDR_S3	Slot:4	ADJ FB=0.5V	ON	ON	TBD	TBD
VCC_SYSIN	RK806-1_BUCK7	2.5A	VCC_2V0_PLDO_S3	Slot:1	2.0V	ON	ON	TBD	TBD
VCC_SYSIN	RK806-1_BUCK8	2.5A	VCC_3V3_S3	Slot:6	3.3V	ON	ON	TBD	TBD
VCC_SYSIN	RK806-1_BUCK9	2.5A	VDDQ_DDR_S0	Slot:6	ADJ FB=0.5V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_BUCK10	2.5A	VCC_1V8_S3	Slot:3	1.8V	ON	ON	TBD	TBD
VCC_2V0_PLDO_S3	RK806-1_PLDO1	0.5A	VCC_1V8_S0	Slot:3	1.8V	ON	OFF	TBD	TBD
	RK806-1_PLDO2	0.3A	VCCA_1V8_S0	Slot:3	1.8V	ON	OFF	TBD	TBD
	RK806-1_PLDO3	0.3A	VDDA_1V2_S0	Slot:4	1.2V	ON	OFF	TBD	TBD
VCC_SYSIN	RK806-1_PLDO4	0.5A	VCCA_3V3_S0	Slot:6	3.3V	ON	OFF	TBD	TBD
	RK806-1_PLDO5	0.3A	VCCIO_SD_S0	Slot:6	3.3V	ON	OFF	TBD	TBD
	RK806-1_PLDO6	0.3A	VCCA1V8_PLDO6_S3	Slot:3	1.8V	ON	ON	TBD	TBD
VCC_1V1_NLDO_S3	RK806-1_NLDO1	0.3A	VDD_0V75_S3	Slot:2	0.75V	ON	ON	TBD	TBD
	RK806-1_NLDO2	0.3A	VDDA_DDR_PLL_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
	RK806-1_NLDO3	0.5A	VDDA_0V75_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC_1V1_NLDO_S3	RK806-1_NLDO4	0.5A	VDDA_0V85_S0	Slot:2	0.85V	ON	OFF	TBD	TBD
	RK806-1_NLDO5	0.3A	VDD_0V75_S0	Slot:2	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	BUCK_RK860-2	6A	VDD_CPU_BIG0_S0	Slot:6A	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	BUCK_RK860-3	6A	VDD_CPU_BIG1_S0	Slot:6A	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	BUCK_RK860-2	6A	VDD_NPU_S0	Slot:6A	0.75V	ON	OFF	TBD	TBD
VCC_SYSIN	EXT BUCK	2A	VCC_1V1_NLDO_S3	Slot:1	1.1V	ON	ON	TBD	TBD
VCC_SYSIN	EXT BUCK	2A	VDD2L_0V9_DDR_S3	Slot:5	0.9V	ON	ON	TBD	TBD
VCC_SYSIN	EXT BUCK	2.5A	VCC_3V3_SD_S0	Slot:6A	3.3V	ON	OFF	TBD	TBD
VCC_SYSIN	EXT_BUCK or LDO	2A	VCC_1V2_CAM_S0	OFF	1.2V	OFF	OFF	TBD	TBD
VCC_SYSIN	LDO	0.5A	VCC_1V8_CAM_S0	OFF	1.8V	OFF	OFF	TBD	TBD
VCC_SYSIN	LDO	0.5A	VCC_2V8_CAM_S0	OFF	2.8V	OFF	OFF	TBD	TBD

IO Power Domain Map

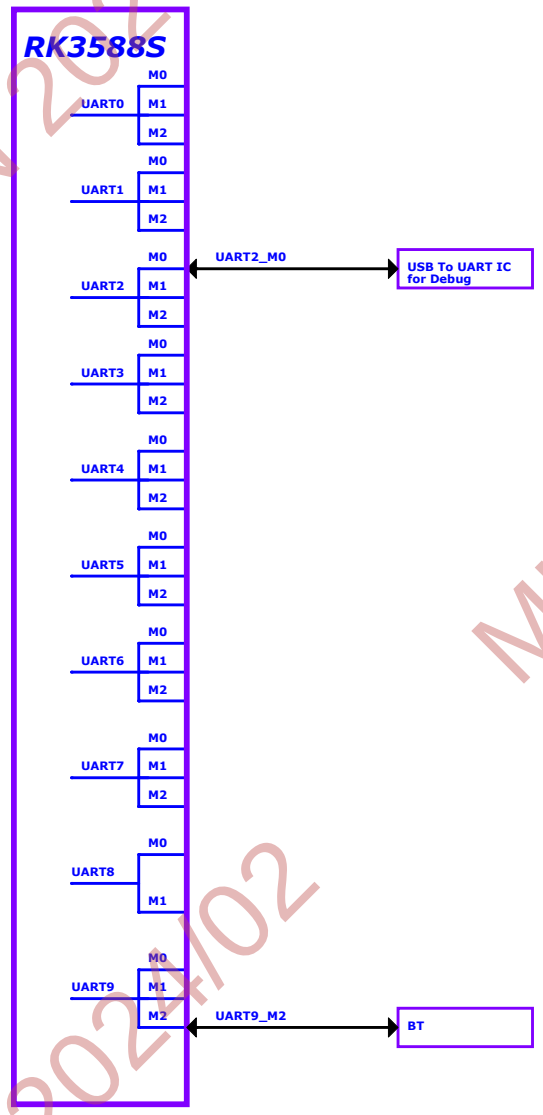
IO Domain	Pin Num	Support IO Voltage	Supply Power Pin Name	Power Source	Operating Voltage
PMUIO1	Pin N36 N37	1.8V Only	PMUIO1_1V8	VCC_1V8_S3	1.8V
PMUIO2	Pin V37 Y37 Pin V35 V36	1.8V or 3.3V	PMUIO2_1V8 PMUIO2_	VCC_1V8_S3 VCC_1V8_S3	1.8V 1.8V
EMMCIO	Pin AC35 Pin AC36	1.8V Only	EMMCIO_1V8	VCC_1V8_S0	1.8V
VCCIO1	Pin H31	1.8V Only	VCCIO1_1V8	VCC_1V8_S0	1.8V
VCCIO2	Pin AK11 Pin AK10	1.8V or 3.3V	VCCIO2_1V8 VCCIO2_	VCC_1V8_S0 VCC_IO_SD	1.8V 1.8V/3.3V
VCCIO4	Pin G27 G28 Pin G31	1.8V or 3.3V	VCCIO4_1V8 VCCIO4_	VCC_1V8_S0 VCC_3V3_S0	1.8V 1.8V
VCCIO5	Pin AF35 AF36 Pin AC33 AC34	1.8V or 3.3V	VCCIO5_1V8 VCCIO5_	VCC_1V8_S0 VCC_1V8_S0	1.8V 1.8V
VCCIO6	Pin AJ34 Pin AL33 AM33	1.8V or 3.3V	VCCIO6_1V8 VCCIO6_	VCC_1V8_S0 VCC_3V3_S0	1.8V 3.3V

MURAT IRKAN 2024/02

I2C MAP



UART MAP



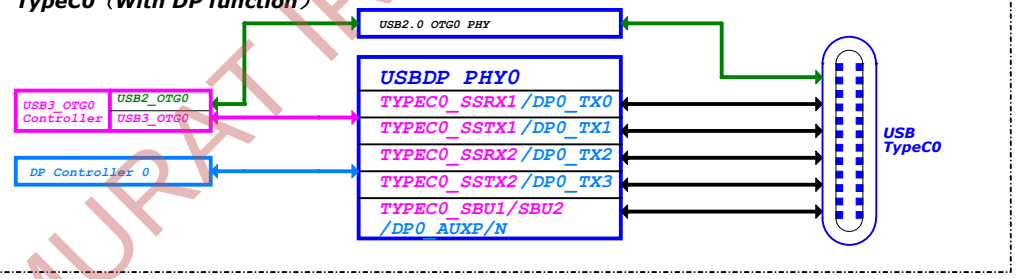
MURAT IRKAN 2024/02

USB Controller Configure Table

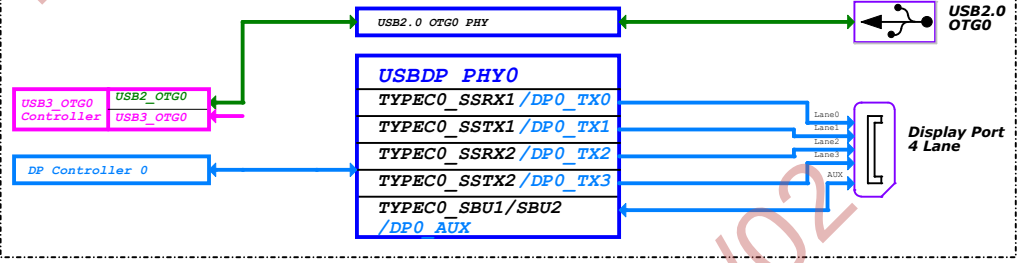
Controller Name	Pin Name	Type-C Function	DPx4Lane+USB2.0 OTG		USB30 OTG+DPx2Lane Function		USB20 OTG+DPx2Lane Function		USB20 OTG+DPx4Lane Function	
			OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2	OPTION1	OPTION2
USB30 OTG0 Device or Host	TYPEC0_SBU1/DP0_AUXP TYPEC0_SBU2/DP0_AUXN	TYPEC0_SBU1 TYPEC0_SBU2	DP0_AUXP DP0_AUXN	DP0_AUXP DP0_AUXN	DP0_AUXP DP0_AUXN	DP0_AUXP DP0_AUXN	DP0_AUXP DP0_AUXN	DP0_AUXP DP0_AUXN	DP0_AUXP DP0_AUXN	DP0_AUXP DP0_AUXN
	TYPEC0_SSRX1P/DP0_TX0P TYPEC0_SSRX1N/DP0_TX0N	TYPEC0_SSRX1P TYPEC0_SSRX1N	DP0_TX0P DP0_TX0N	DP0_TX2P DP0_TX2N	TYPEC0_SSRX1P TYPEC0_SSRX1N	DP0_TX0P DP0_TX0N	DP0_TX0P DP0_TX0N		DP0_TX0P DP0_TX0N	DP0_TX2P DP0_TX2N
	TYPEC0_SSTX1P/DP0_TX1P TYPEC0_SSTX1N/DP0_TX1N	TYPEC0_SSTX1P TYPEC0_SSTX1N	DP0_TX1P DP0_TX1N	DP0_TX3P DP0_TX3N	TYPEC0_SSTX1P TYPEC0_SSTX1N	DP0_TX1P DP0_TX1N	DP0_TX1P DP0_TX1N		DP0_TX1P DP0_TX1N	DP0_TX3P DP0_TX3N
	TYPEC0_SSRX2P/DP0_TX2P TYPEC0_SSRX2N/DP0_TX2N	TYPEC0_SSRX2P TYPEC0_SSRX2N	DP0_TX2P DP0_TX2N	DP0_TX0P DP0_TX0N	DP0_TX2P DP0_TX2N	TYPEC0_SSRX2P TYPEC0_SSRX2N		DP0_TX2P DP0_TX2N	DP0_TX2P DP0_TX2N	DP0_TX0P DP0_TX0N
	TYPEC0_SSTX2P/DP0_TX3P TYPEC0_SSTX2N/DP0_TX3N	TYPEC0_SSTX2P TYPEC0_SSTX2N	DP0_TX3P DP0_TX3N	DP0_TX1P DP0_TX1N	DP0_TX3P DP0_TX3N	TYPEC0_SSTX2P TYPEC0_SSTX2N		DP0_TX3P DP0_TX3N	DP0_TX3P DP0_TX3N	DP0_TX1P DP0_TX1N
USB20 OTG0 Device or Host	TYPEC0_USB20_OTG_DP TYPEC0_USB20_OTG_DM	TYPEC0_USB20_OTG_DP TYPEC0_USB20_OTG_DM	TYPEC0_USB20_OTG_DP TYPEC0_USB20_OTG_DM		TYPEC0_USB20_OTG_DP TYPEC0_USB20_OTG_DM		TYPEC0_USB20_OTG_DP TYPEC0_USB20_OTG_DM		TYPEC0_USB20_OTG_DP TYPEC0_USB20_OTG_DM	TYPEC0_USB20_OTG_DP TYPEC0_USB20_OTG_DM
			OPTION1 USB30 HOST		OPTION2 USB30 HOST					
USB30 OTG2 Device or Host	PCIE20_2_TXP/SATA30_2_TXP/USB30_2_SSTXP PCIE20_2_TXN/SATA30_2_TXN/USB30_2_SSTXN		USB30_2_SSTXP USB30_2_SSTXN		USB30_2_SSTXP USB30_2_SSTXN					
	PCIE20_2_RXP/SATA30_2_RXP/USB30_2_SSRXP PCIE20_2_RXN/SATA30_2_RXN/USB30_2_SSRXN		USB30_2_SSRXP USB30_2_SSRXN		USB30_2_SSRXP USB30_2_SSRXN					
USB20 HOST0	USB20_HOST0_DP USB20_HOST0_DM		USB20_HOST0_DP USB20_HOST0_DM							
USB20 HOST1	USB20_HOST1_DP USB20_HOST1_DM				USB20_HOST1_DP USB20_HOST1_DM					

Note:
DP Lane swap enable
0:Lane0/1/2/3 TxData mapping to Lane0/1/2/3 TXDP/N
1:Lane0/1/2/3 TxData mapping to Lane2/3/0/1 TXDP/N

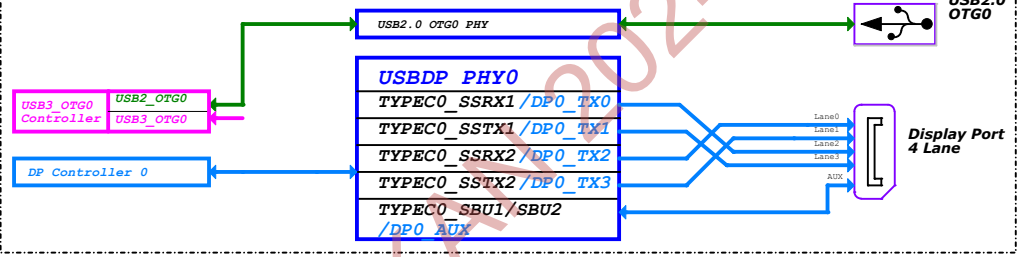
Config0:
TypeC0（With DP function）



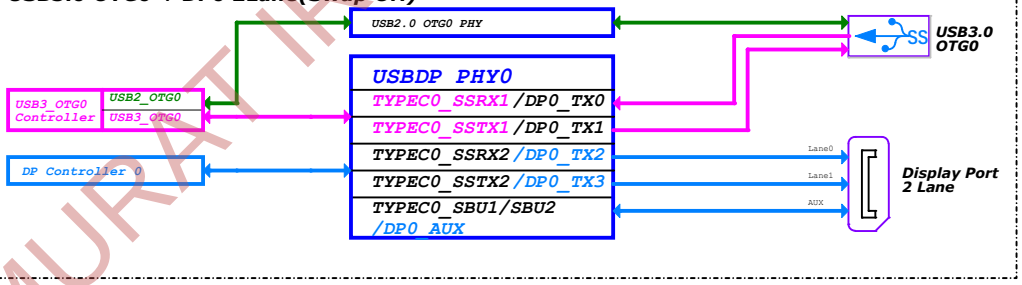
Config1:
USB2.0 OTG0 + DP0 4Lane(Swap OFF)



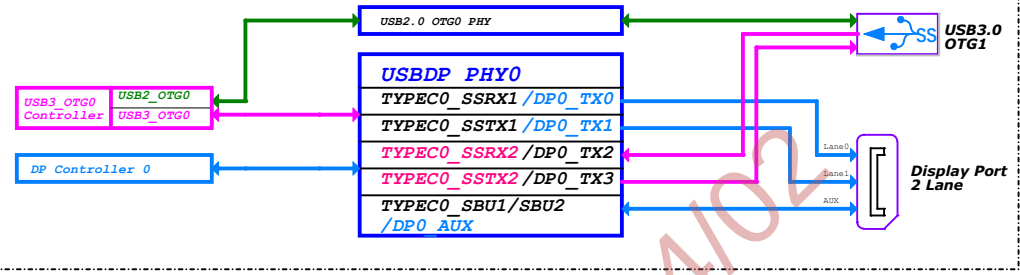
Config2:
USB2.0 OTG0 + DP0 4Lane(Swap ON)



Config3:(Default)
USB3.0 OTG0 + DP0 2Lane(Swap ON)

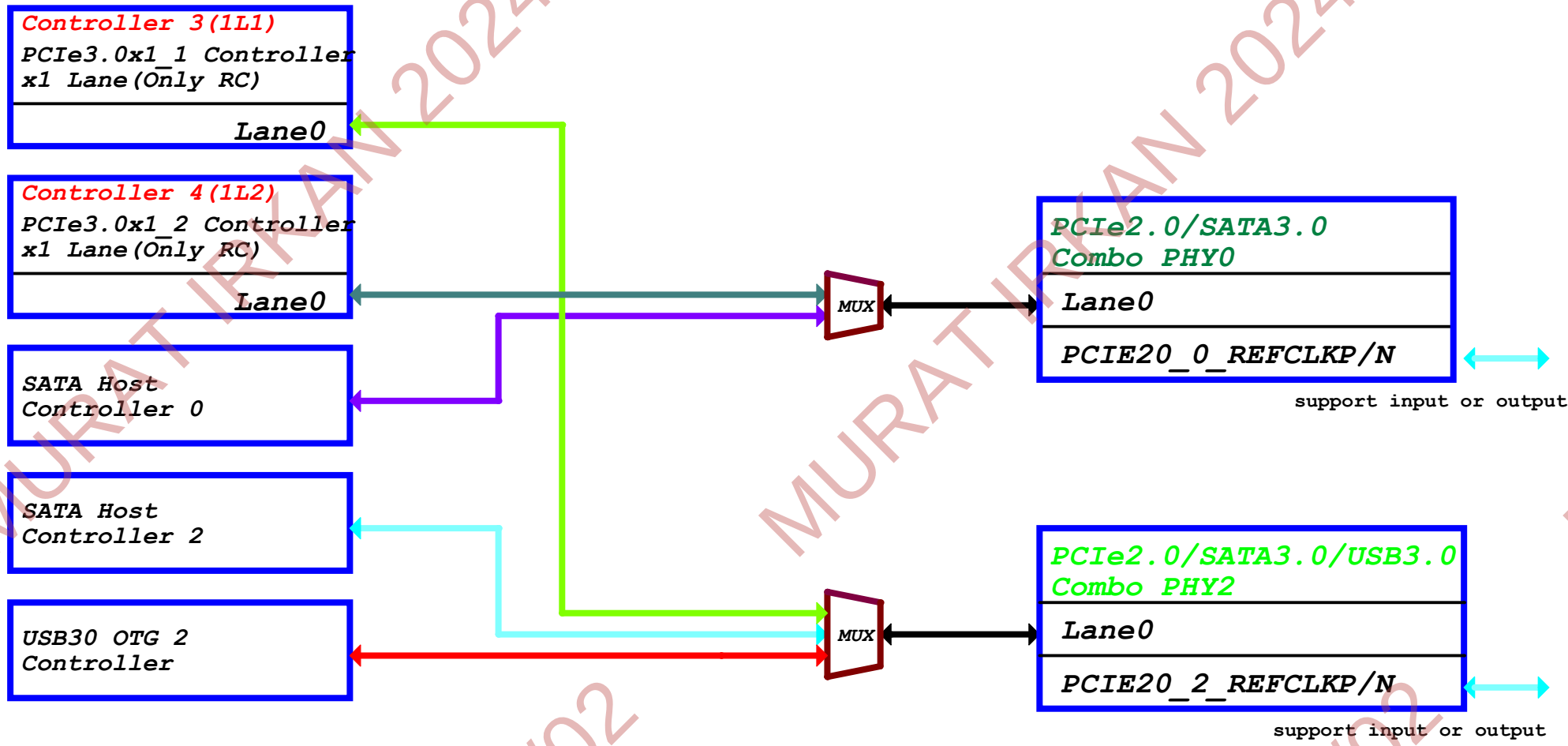


Config4:
USB3.0 OTG0 + DP0 2Lane(Swap OFF)



MURAT IRKAN 2024/02

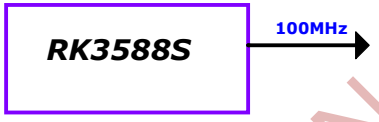
PCIe/SATA Connecter Diagram



PCIe Controller Configure Table

Controller Name	Data & Clk Lane Configure		Control GPIO
	CLK LANE	DATA LANE	
PCIE20X1_1 RC	PCIE20_2_REFCLKP PCIE20_2_REFCLKN	PCIE20_2_TX PCIE20_2_RX	PCIE20X1_1_CLKREQ_M* PCIE20X1_1_WAKEN_M* PCIE20X1_1_PERSTN_M* PCIE20X1_1_BUTTON_RSTN
PCIE20X1_2 RC	PCIE20_0_REFCLKP PCIE20_0_REFCLKN	PCIE20_0_TX PCIE20_0_RX	PCIE20X1_2_CLKREQ_M* PCIE20X1_2_WAKEN_M* PCIE20X1_2_PERSTN_M* PCIE20X1_2_BUTTON_RSTN

PCIe2.0 REFCLK



Note:

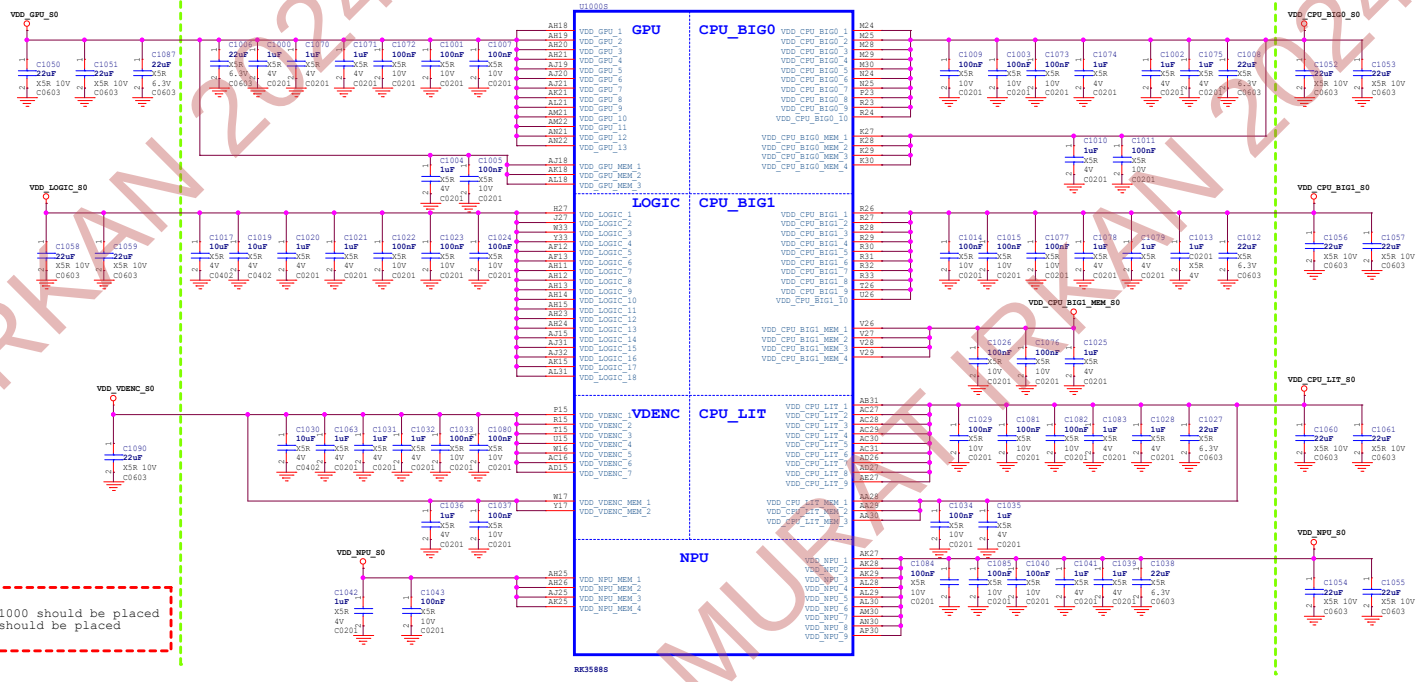
PCIE20_*_REFCLKP/N is output or input gpio

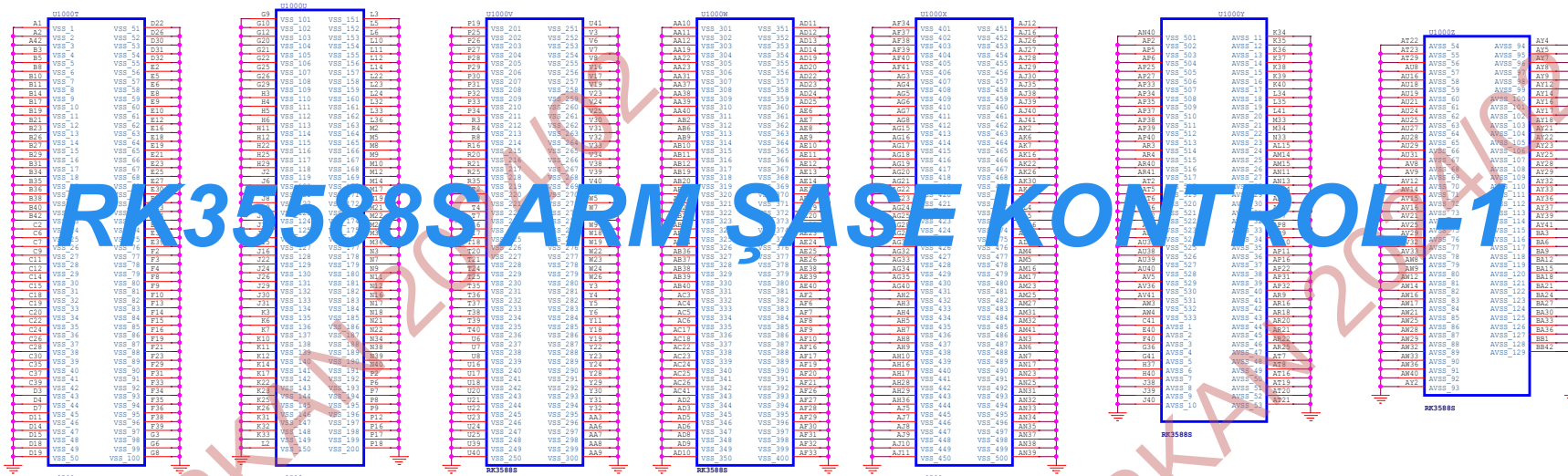
M*=Mean to M0 or M1 or M2,It's the same source,Just multiplex to M0 or M1 or M2,Only use one at the same time.

MURAT IRKAN 2024/02

RK3588S(Power&Gnd)

WINDOWS 11 / LINUX ARM NOTEBOOK -ROCKCHIP RK3588S





RK3588S ARM ŞASE KONTROL-1

MURAT IRKAN 2024/02

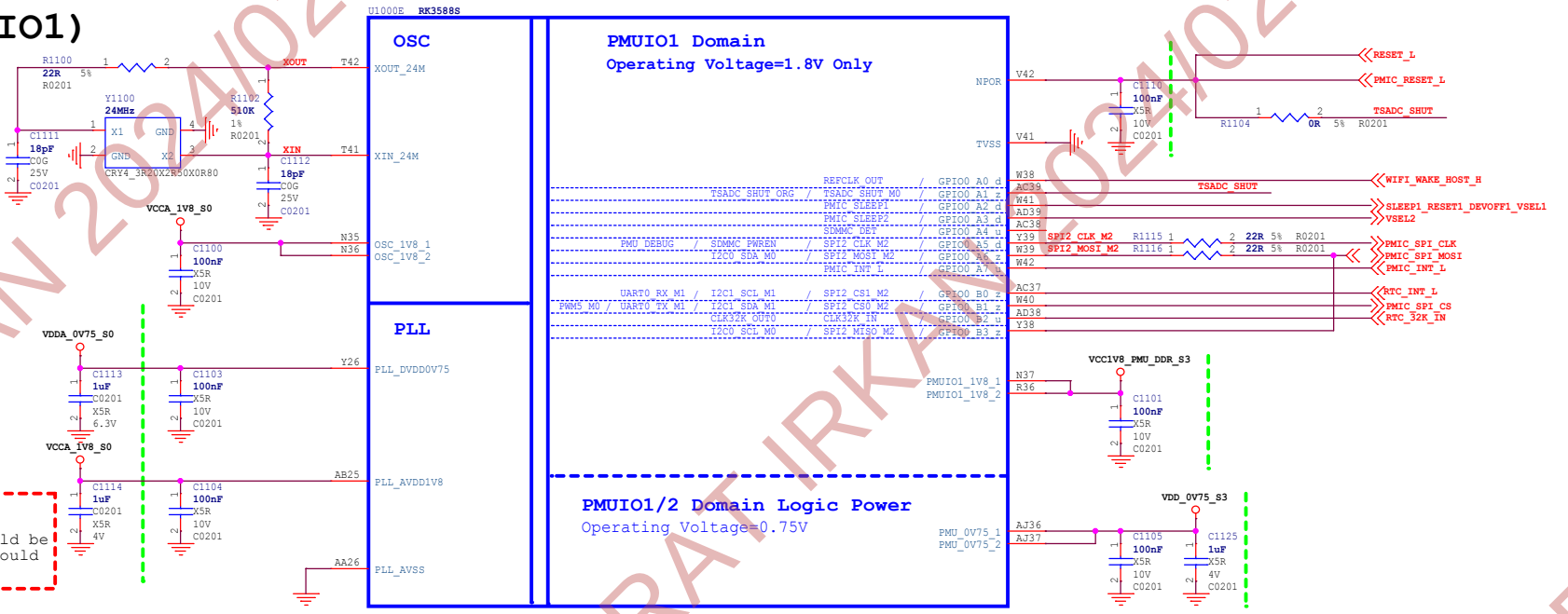
RK3588S(OSC/PLL/PMUIO1)

Note:
Adjusted the load capacitance according to the crystal specification

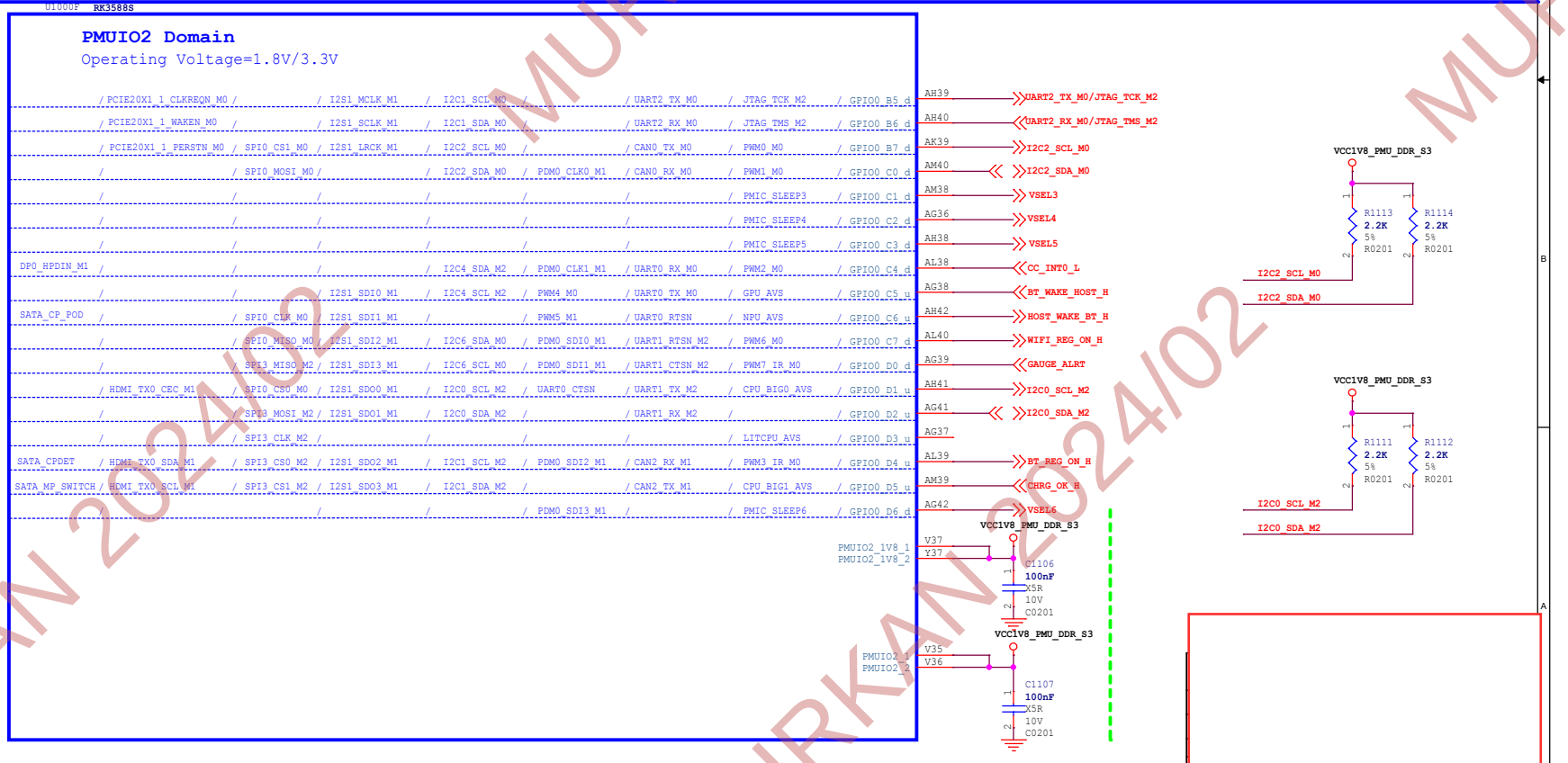
The CL is the load capacitance of the crystal that is recommended by the crystal vendors to obtain target clock frequency.

CL={CL1*CL2/(CL1+CL2))}+PCB strays
Total CL<=12pF

Note:
The Caps between green line and U1000 should be placed under the U1000 package.Other caps should be placed close to the U1000 package



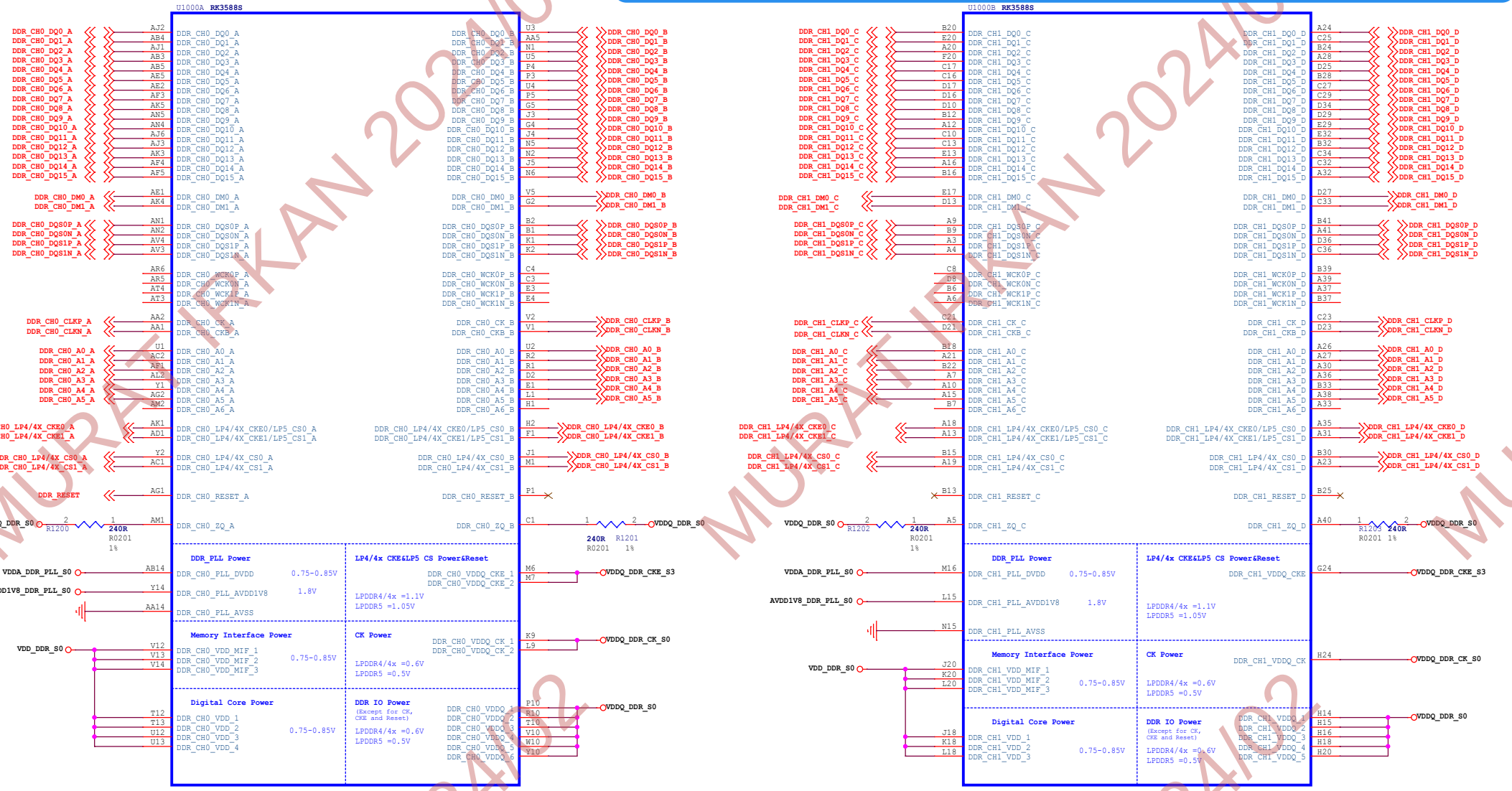
RK3588S(PMUIO2)



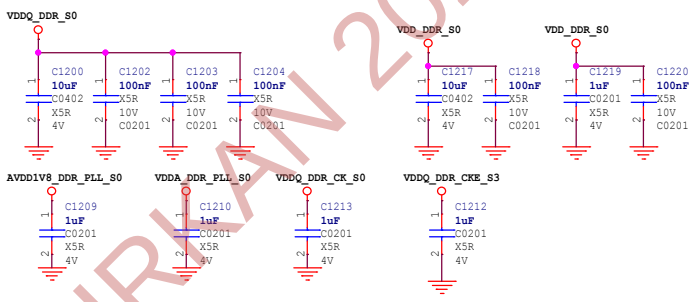
MURAT IRKAN 2024/02

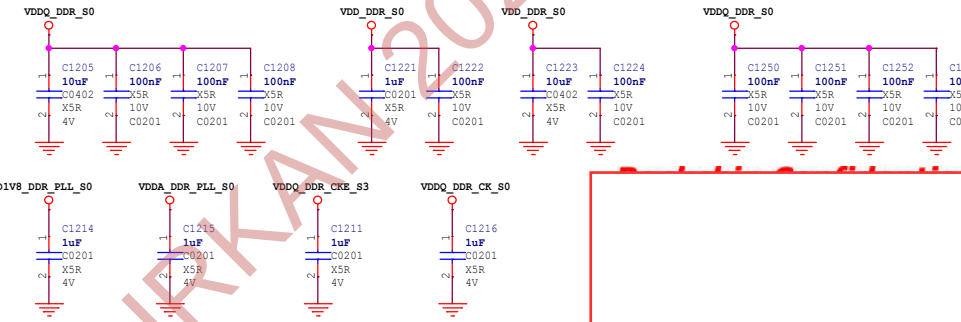
RK3588S(DDR PHY)

LPDDR4-MICRON 32GB



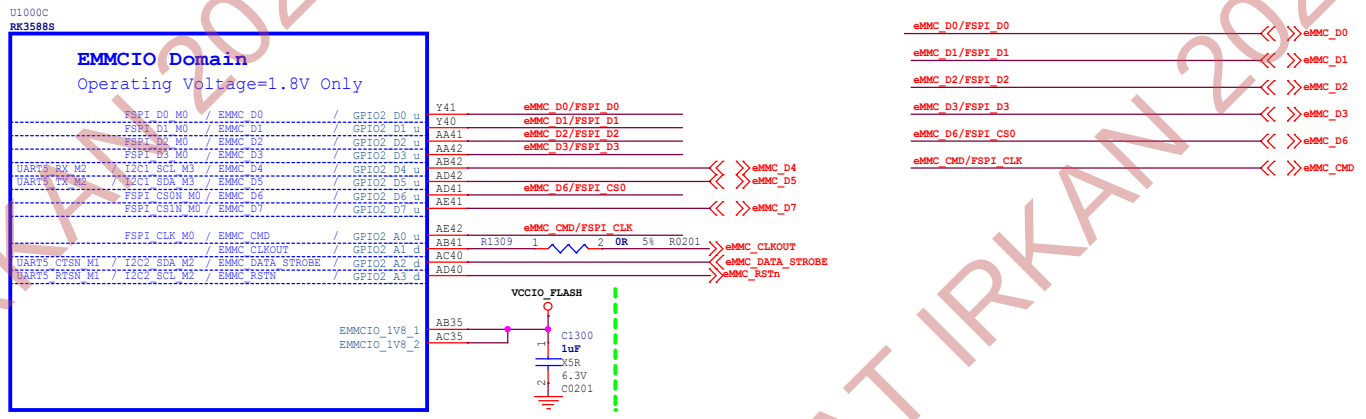
DDR FILTER





MURAT IRKAN 2024/02

RK3588S(EMMCIO Domain)



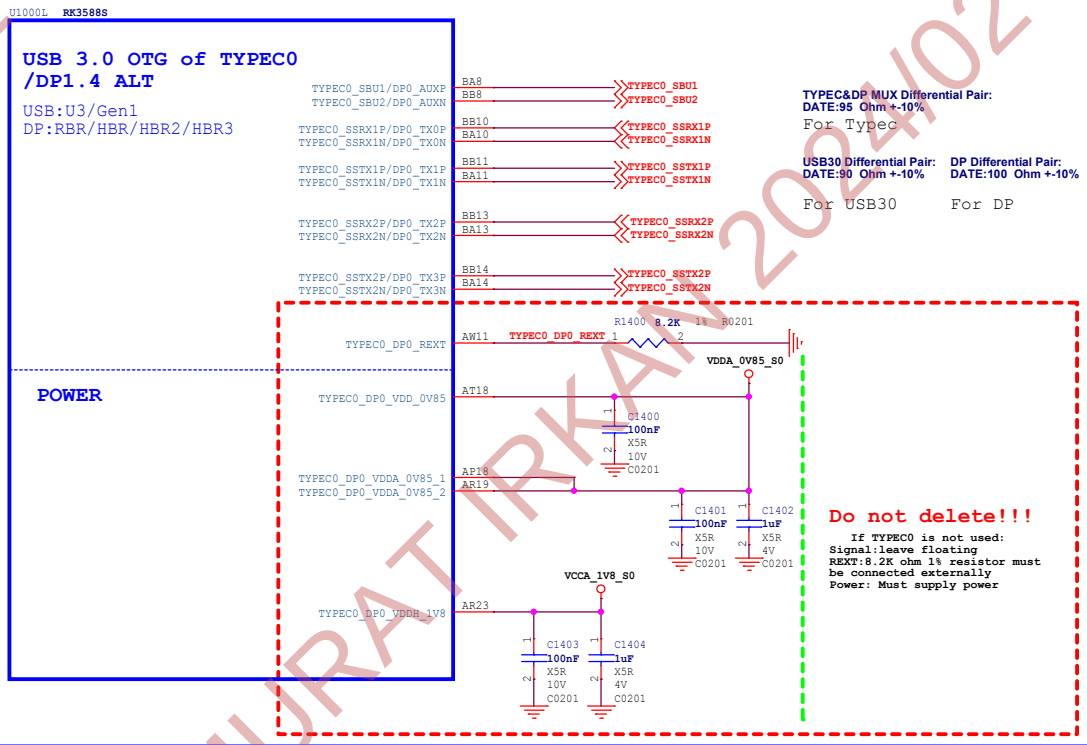
RK3588S(VCCIO2 Domain)



Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

MURAT IRKAN 2024/02

RK3588S(USB3.0/DP1.4)

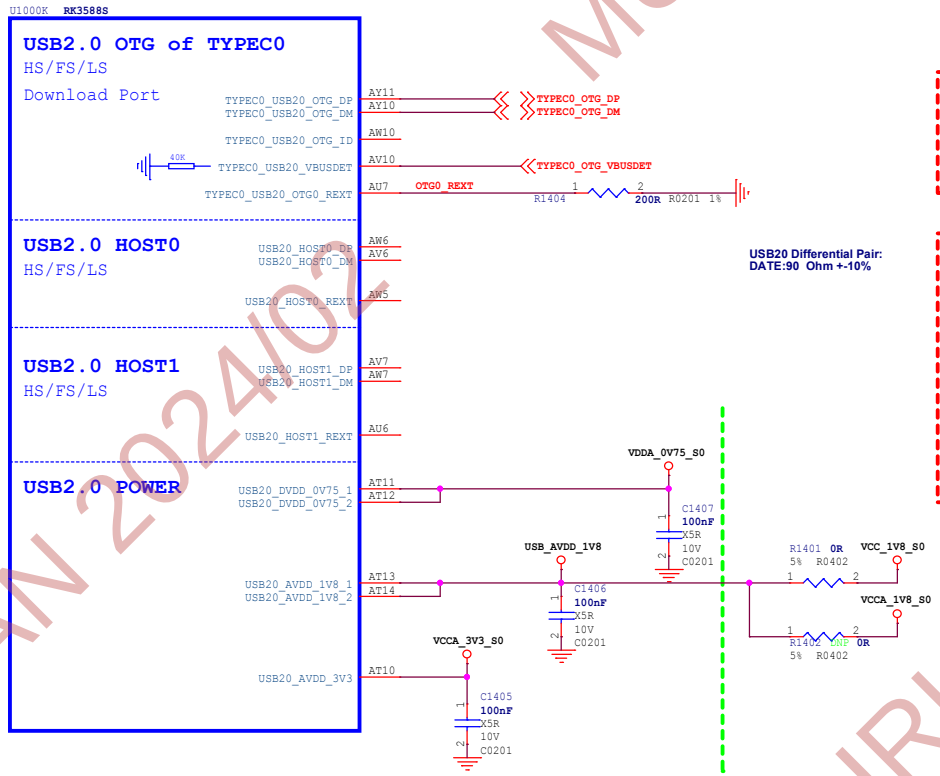


USB30/DP1.4 Alt Mode Configuration

Option1	*DP x4Lane*	*DP_TX_Lane0-3*
Option2	*TYPEC x4Lane*	*SSTX 1P/1N SSTX 2P/2N SSRX 1P/1N SSRX 2P/2N*
Option3	*USB30X2Lane+DPX2Lane*	*USB30:SSTX 1P/1N SSRX 1P/1N DP:Lane2 Lane3*
Option4	*USB30X2Lane+DPX2Lane*	*USB30:SSTX 2P/2N SSRX 2P/2N DP:Lane0 Lane1*

DP Lane
Swap Off:
Lane0/1/2/3_TXdata mapping to Lane0/1/2/3_TXDP/N
Swap On:
Lane0/1/2/3_TXdata mapping to Lane2/3/0/1_TXDP/N

RK3588S(USB2.0)



Note:
The USB20_VBUSDET pin internal has a pull-down resistance(40K ohm) to ground,The resistance creates a voltage with the external series 24K ohm resistor.The VBUSDETpin voltage range <=3.3V.

Note:
TYPEC0_USB20_OTG:

DP/DM:Must used for download
ID:According to demand,if not used,Leave floating
VBUSDET:Must provide
REXT:200ohm 1% resistor must be connected externally
Power: Must supply power

USB20_HOST0/USB20_HOST1:

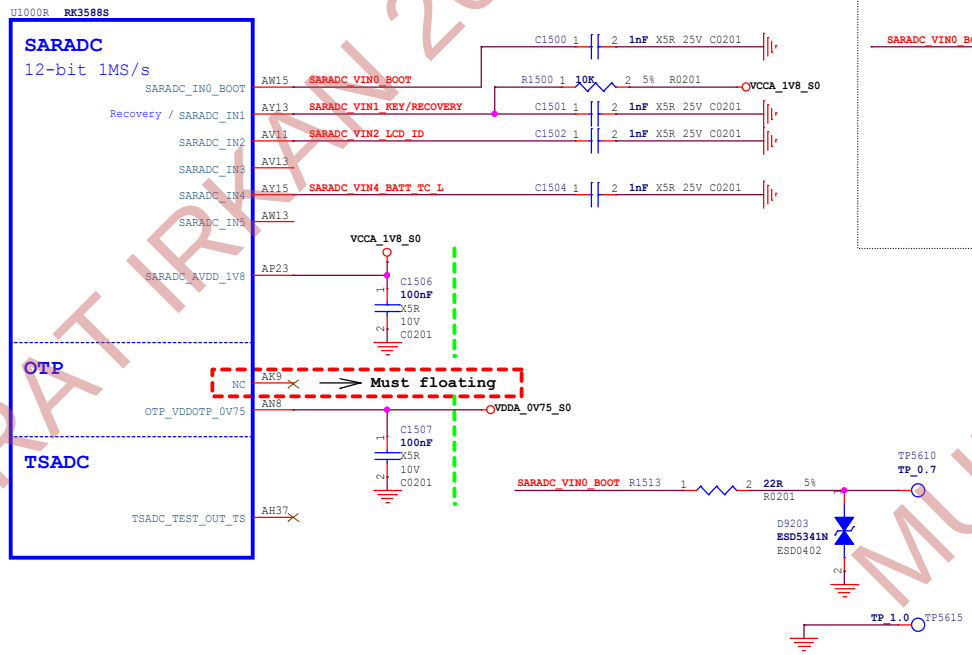
If not used:
DP/DM:Leave floating
REXT:Leave floating

Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

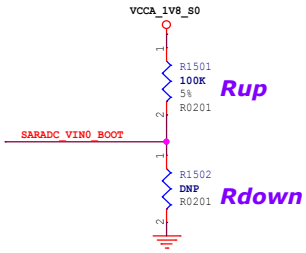
MURAT IRKAN 2024/02

RK3588S(SARADC/OTP/TSADC)





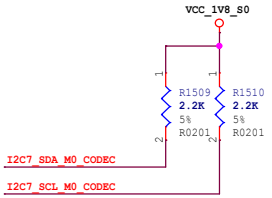
BOOT MODE CONFIG



Item	Rup	Rdown	ADC	BOOT MODE(saradc_in5)
LEVEL1	DNP	100K	0	USB (Maskrom mode)
LEVEL2	100K	20K	682	SD Card-USB
LEVEL3	100K	51K	1365	EMMC-USB
LEVEL4	100K	100K	2047	FSPI M0-USB
LEVEL5	100K	200K	2730	FSPI M1-USB
LEVEL6	100K	499K	3412	FSPI M2-USB
LEVEL7	100K	DNP	4095	FSPI_M2-FSPI_M0-EMMC -SD Card-USB

RK3588S(VCCIO1 Domain)

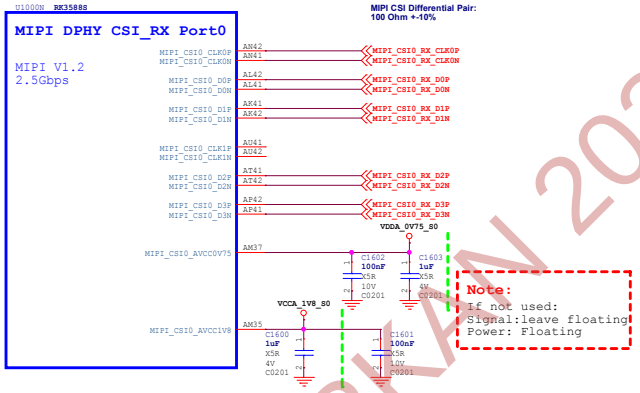




Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

MURAT IRKAN 2024/02

RK3588S(MIPI_DPHY CSI0 RX)

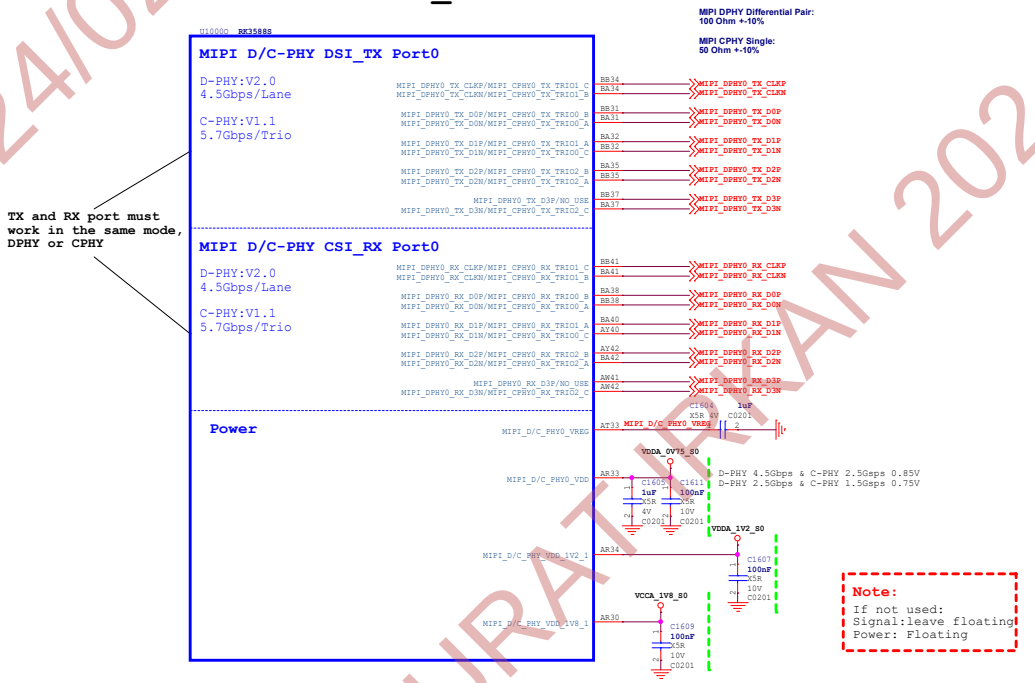


Option1	*Sensor1 x4Lane*	*MIPI_CSI_RX_D0-3* *MIPI_CSI_RX_CLK0*
Option2	*Sensor1 x2Lane* + *Sensor2 x2Lane*	*MIPI_CSI_RX_D0-1* *MIPI_CSI_RX_CLK0* *MIPI_CSI_RX_D2-3* *MIPI_CSI_RX_CLK1*

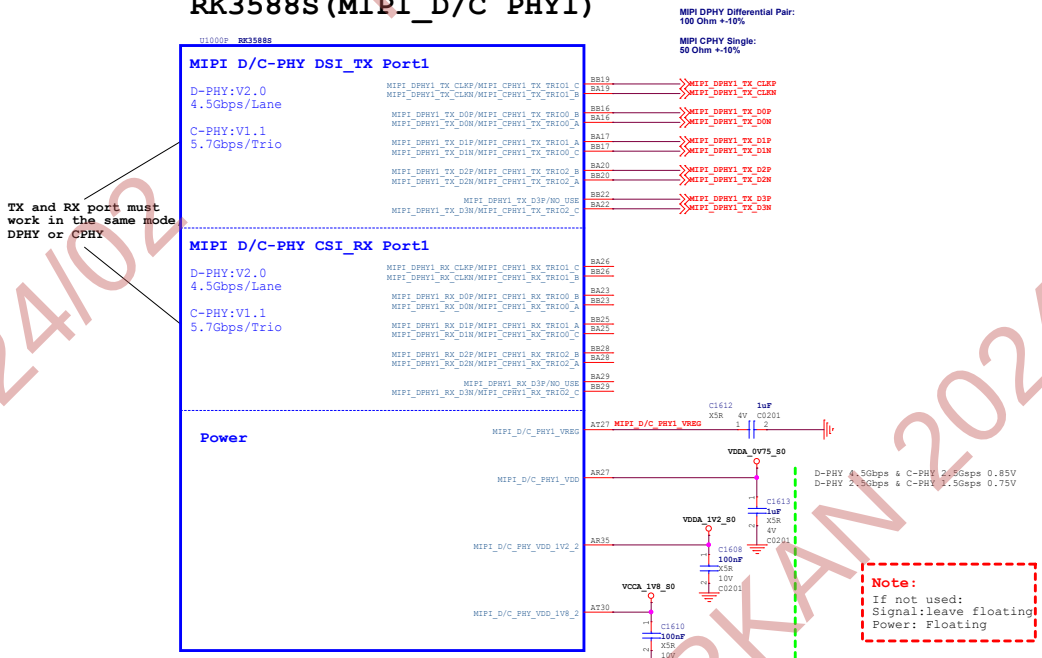
Note:
When in single clock lane mode, CLK0P/0N is the clock lane from Data lane0 to Data lane3,but clock lane1 is invalid; In dual clock lanes mode,CLK0P/0N is the clock lane of Data lane0 and Data lane1,while CLK1P/1N is the clock lane of Data lane2 and Data lane3.

Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

RK3588S(MIPI_D/C PHY0)



RK3588S(MIPI_D/C PHY1)



MURAT IRKAN 2024/02

RK3588S(HDMI2.1 TX/eDP1.3 TX)

Note:
The HDMI2.1 trace length is less than 100mm.
The HDMI2.1 differential trace impedance is 100 OHM.

eDP TX
100 Ohm ±10%

HDMI TX
100 Ohm ±10%

U1000Q

HDMI TX/eDP1.3 MUX Port0

HDMI:V2.1 12Gbps
eDP: V1.3 5.4Gbps

HDMI_TX0_D0P/EDP_TX0_D0P	BB4
HDMI_TX0_D0N/EDP_TX0_D0N	BA4
HDMI_TX0_D1P/EDP_TX0_D1P	BA5
HDMI_TX0_D1N/EDP_TX0_D1N	BB5
HDMI_TX0_D2P/EDP_TX0_D2P	BB7
HDMI_TX0_D2N/EDP_TX0_D2N	BA7
HDMI_TX0_D3P/EDP_TX0_D3P	BA2
HDMI_TX0_D3N/EDP_TX0_D3N	BB2
HDMI_TX0_SBDP/EDP_TX0_AUXP	BA1
HDMI_TX0_SBDN/EDP_TX0_AUXN	AY1
HDMI/eDP_TX0_REXT	AY3

HDMI/eDP_TX0_REXT R1708 1 2 8.2K 1% R0201 VDDA_0V75_S0

POWER

HDMI/EDP_TX0_VDD_0V75_1	AM13
HDMI/EDP_TX0_VDD_0V75_2	AN12
HDMI/EDP_TX0_AVDD_0V75	AN10
HDMI/EDP_TX0_VDD_IO_1V8	AK12
HDMI/EDP_TX0_VDD_CMN_1V8	AL14

C1710 100nF X5R 10V C0201
C1711 4.7uF X5R 6.3V C0402
C1712 100nF X5R 10V C0201
C1725 1uF X5R 4V C0201
C1713 100nF X5R 10V C0201
C1714 4.7uF X5R 6.3V C0402

VCCA_1V8_S0

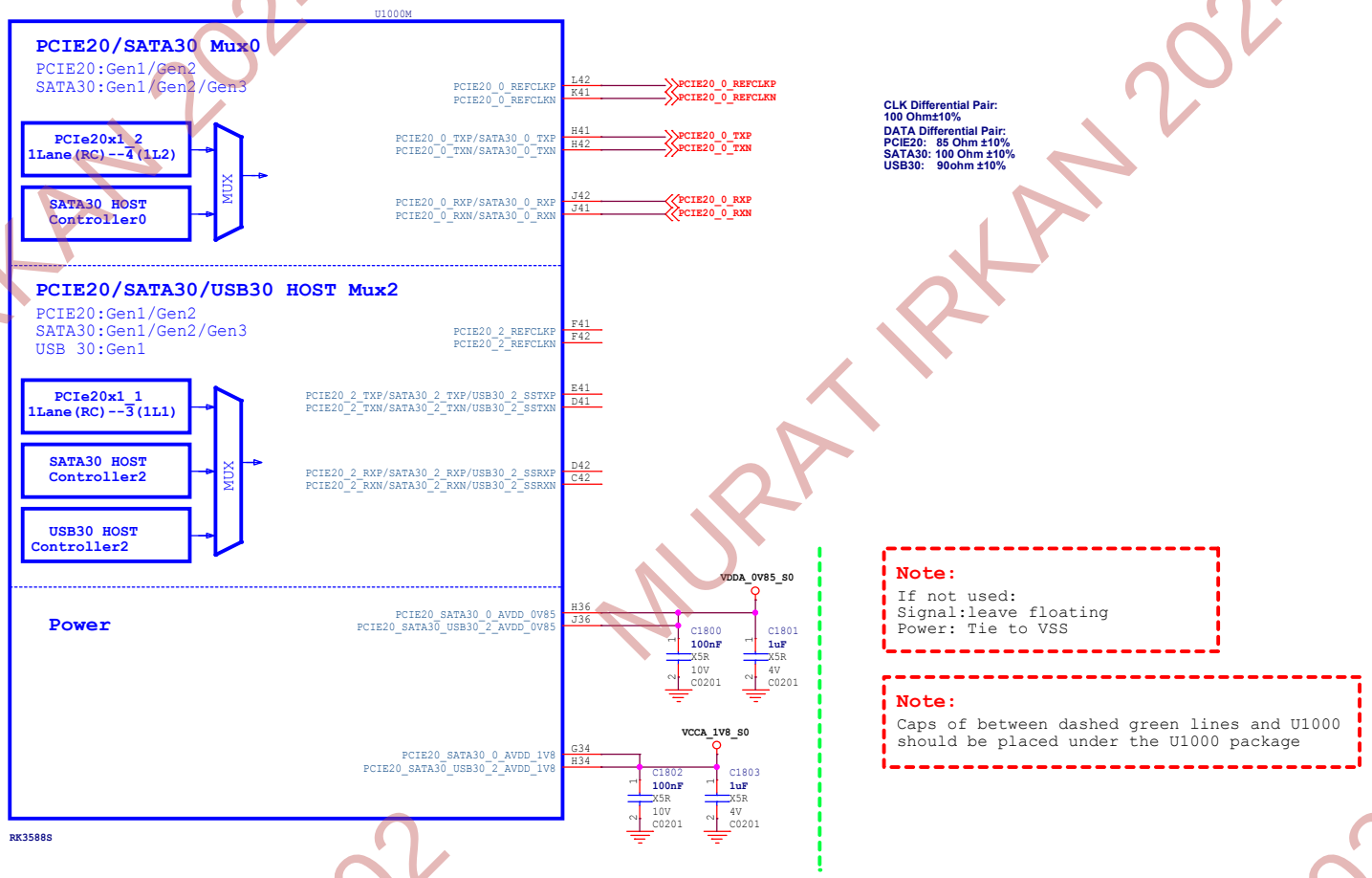
RK3588S

Note:
If not used:
Signal:leave floating
Power: Floating or tie to VSS

Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

MURAT IRKAN 2024/02

RK3588S(PCIE20/SATA30/USB30)



CLK Differential Pair:
100 Ohm±10%
DATA Differential Pair:
PCIE20: 85 Ohm ±10%
SATA30: 100 Ohm ±10%
USB30: 90ohm ±10%

Note:
If not used:
Signal:leave floating
Power: Tie to VSS

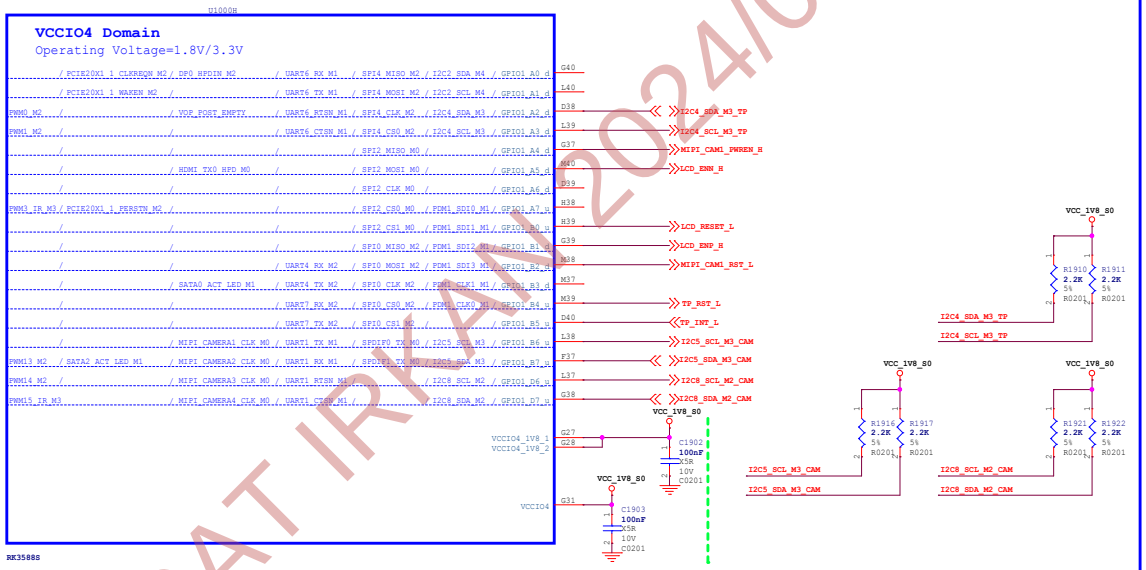
Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package

PCIe2.0 PHY

Controller Name	Data & Clk Lane Configure		Control GPIO
	CLK LANE	DATA LANE	
PCIE20X1_1 RC	PCIE20_2_REFCLKP PCIE20_2_REFCLKN	PCIE20_2_TX PCIE20_2_RX	PCIE20X1_1_CLKREQ_M* PCIE20X1_1_WAKEN_M* PCIE20X1_1_PERSTN_M* PCIE20X1_1_BUTTON_RSTN
PCIE20X1_2 RC	PCIE20_0_REFCLKP PCIE20_0_REFCLKN	PCIE20_0_TX PCIE20_0_RX	PCIE20X1_2_CLKREQ_M* PCIE20X1_2_WAKEN_M* PCIE20X1_2_PERSTN_M* PCIE20X1_2_BUTTON_RSTN

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RK3588S(VCCIO4 Domain)

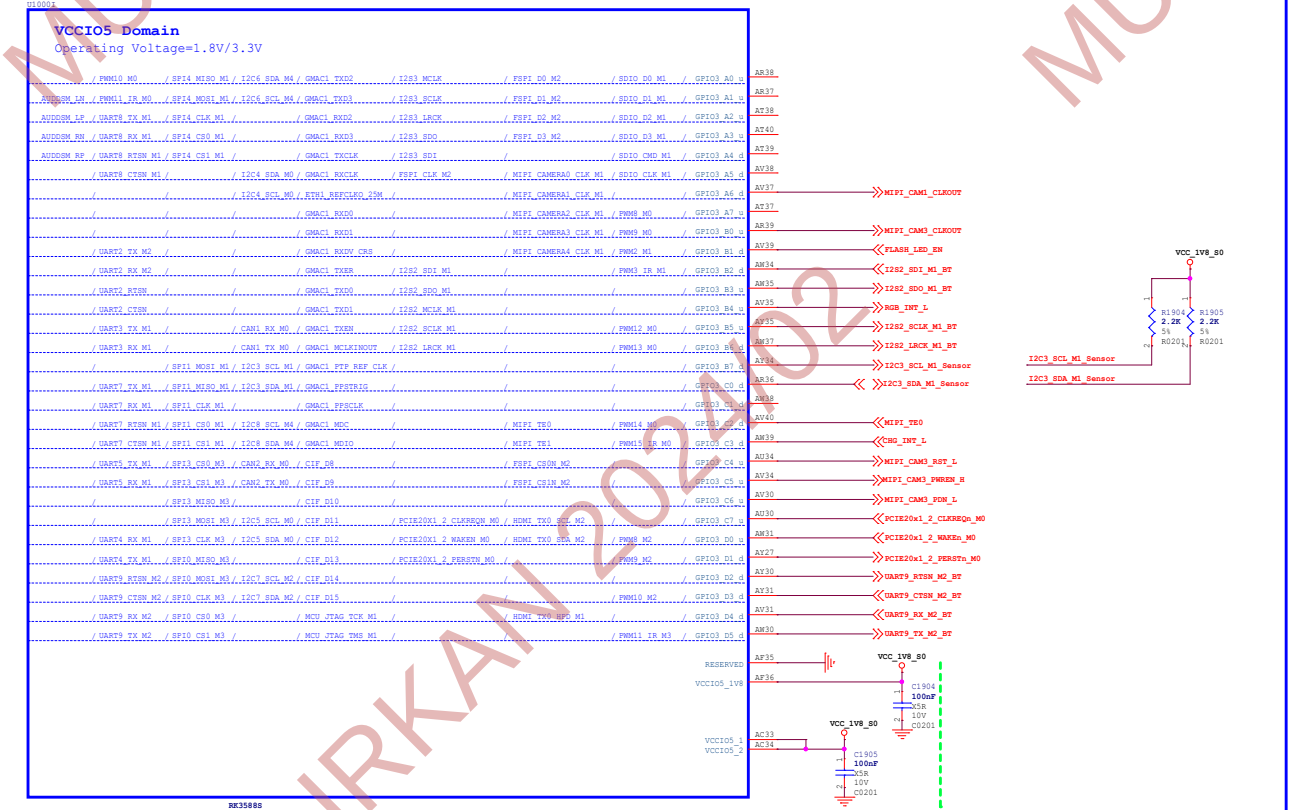


RK3588S(VCCIO6 Domain)

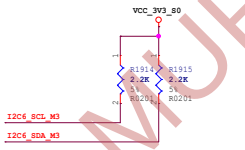


Note:
BT1120 Only Support Output

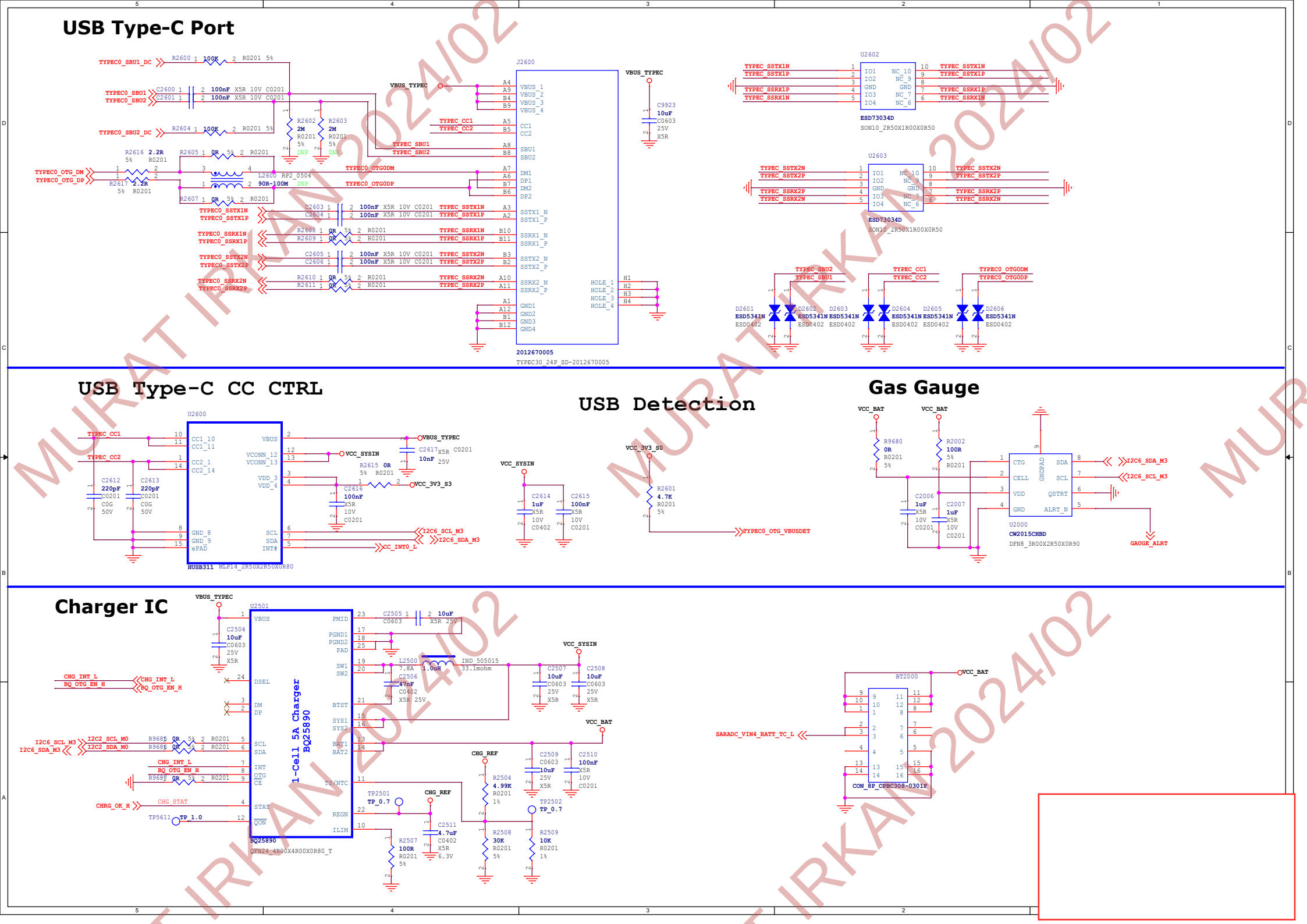
RK3588S(VCCIO5 Domain)



Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package



MURAT IRKAN 2024/02


USB Type-C Port
TYPEC0_SBU1_DC
TYPEC0_SBU2_DC
TYPEC0_SBU1
TYPEC0_SBU2
TYPEC0_OTG_DM
TYPEC0_OTG_DP
TYPEC0_SSTX1N
TYPEC0_SSTX1P
TYPEC0_SSRX1N
TYPEC0_SSRX1P
TYPEC0_SSTX2N
TYPEC0_SSTX2P
TYPEC0_SSRX2N
TYPEC0_SSRX2P
VBUS_TYPEC
J2600
2012670005
TYPEC30_24P_SD-2012670005
U2602
ESD73034D
SON10_2R50X1R00X0R50
U2603
D2601
D2602
D2603
D2604
D2605
D2606
ESD5341N
ESD0402
USB Type-C CC CTRL
U2600
HUSB311
MLP14_2R50X2R50X0R80
USB Detection
VCC_3V3_S0
TYPEC0_OTG_VBUSDET
Gas Gauge
VCC_BAT
U2000
CW2015CHBD
DFN8_3R00X2R50X0R90
GAUGE_ALRT
I2C6_SCL_M3
I2C6_SDA_M3
Charger IC
U2501
BQ25890
1-Cell 5A Charger
QFN24_4R00X4R00X0R80_T
CHG_INT_L
BQ_OTG_EN_H
CHG_OK_H
CHG_STAT
VCC_SYSIN
VCC_3V3_S3
CC_INT0_L
BT2000
CON_8P_CPBC308-0301F
SARADC_VIN4_BATT_TC_L

PMIC1 RK806-1

SLEEP1_RESET1_DEVOFF1_VSEL1
VSEL4
VSEL5
PMIC_INT_L
PWRON_L
PMIC_RESET_L
PMIC_EXT_EN_OUT
PMIC_SPI_CLK
PMIC_SPI_CS
PMIC_SPI_MOSI

VCC_SYSIN

IF TVS UNMOUNTED, ESD OR SURGE SHOULD BE DAMAGE THE PMIC!!!

ED2200 AZ5825-01F ESD0402

This device must be mounted.Replacing TVS mode is not recommended, if must, please choose the same specifications
Operating Supply Vltage : 5.5V(5.25-6V)
PeakPulse Current: >10A (tp=8/20uS)
Surge Clamping Voltage: <6.5V

DO NOT DELETE IT!

VSEL5 — R2242 0R — R0201 5% — PMIC_PWR_CTRL2
R2244 DNP 120K R0201

VSEL4 — R2290 0R — R0201 5% — PMIC_PWR_CTRL3
R2291 DNP 120K R0201

PMIC RK806-1 BUCK

Default:0.85V
Low frequency:
0.85V-->0.75V

VDD_DDR_S0 — Feedback from RK3588S

LPDDR4/4x=1.1V	120K	
LPDDR5=1.05V	110K	

VDDQ_DDR_S0

LPDDR4/4x=0.6V	20K	100K
LPDDR5=0.5V	0R	DNP

VCC_3V3_S3

VCC1V8_PMU_DDR_S3 — VCC_1V8_S3_PLDO6

Default:1.8V

VCC_1V8_S3

U2200A

BUCK5 2.5A		BUCK1 6.5A
BUCK6 2.5A		BUCK2 5A
BUCK9 2.5A		BUCK3 5A
BUCK8 2.5A		BUCK4 5A
BUCK10 2.5A		BUCK7 2.5A

RK806-1

VDD_GPU_S0 — Feedback from RK3588S
VDD_CPU_LIT_S0 — Feedback from RK3588S
VDD_LOGIC_S0 — Feedback from RK3588S
VDD_VDENC_S0 — Feedback from RK3588
VCC_2V0_PLDO_S3

PMIC RK806-1 Managerment

U2200C

VCCIO (IIC/SPI)CS — PMIC_SPI_CS
(SDA)MOSI — PMIC_SPI_MOSI
(SCL)CLK — PMIC_SPI_CLK
(PWRCTRL3)SO — PMIC_PWR_CTRL3
PWRCTRL2 — PMIC_PWR_CTRL2
PWRCTRL1 — SLEEP1_RESET1_DEVOFF1_VSEL1

VCCA

SYNC_CLK
SYNC
VDC
PWRON — PWRON_L
(M/S)EXT_EN — PMIC_EXT_EN_OUT
INT — PMIC_INT_L
RESETB — PMIC_RESET_L
ePAD

VBUS_TYPEC — PMIC_VDC — D2600 BAT752C3V0 SOD_123

RK806-1

RK3588S
nPOR
TSADC_SHUT

RK806-1
RESETB

Reset Key

Note:

- Reset Key Control Path
- TSADC_SHUT Control Path
- RK806-1 Control Path

PMIC RK806-1 LDO

U2200B

PLDO1	0.5A	VCC_1V8_S0	Default:1.8V
PLDO2	0.3A	VCCA_1V8_S0	Default:1.8V
PLDO3	0.3A	VDDA_1V2_S0	Default:1.2V
PLDO4	0.5A	VCCA_3V3_S0	Default:3.3V
PLDO5	0.3A	VCCIO_SD_S0	Default:3.3V
NLDO1	0.3A	VDD_0V75_S3	Default:0.75V
NLDO2	0.3A	VDDA_DDR_PLL_S0	Default:0.85V; Low frequency:0.85V-->0.75V
NLDO3	0.5A	VDDA_0V75_S0	Default:0.75V
NLDO4	0.5A	VDDA_0V85_S0	Default:0.85V
NLDO5	0.3A	C2265 2.2uF	Default:0.75V Reserved

VCC_2V0_PLDO_S3 — VCC_SYSIN — VCC_1V1_NLDO_S3 — VCC_1V1_NLDO_S3

PLDO

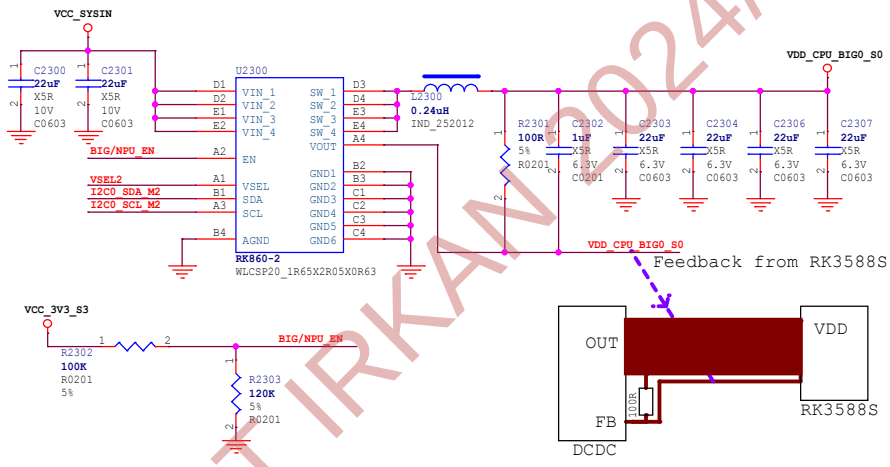
NLDO

RK806-1

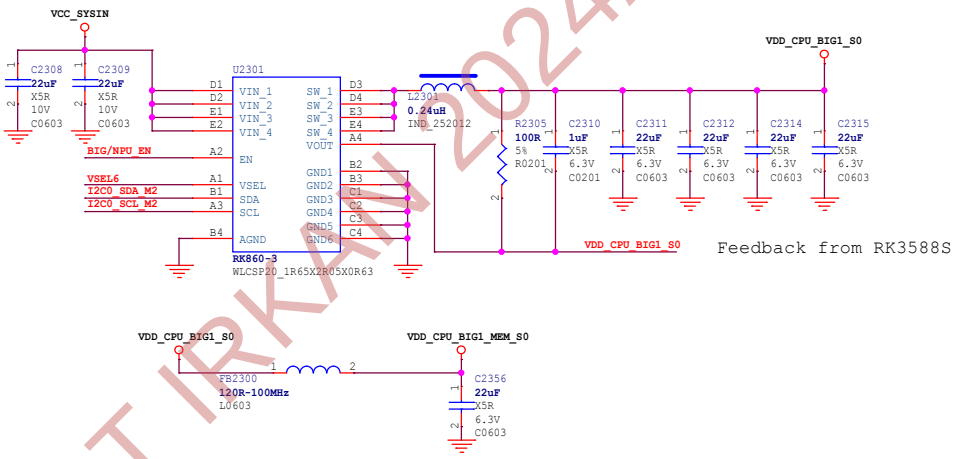
VCCA_1V8_S0 — AVDD1V8_DDR_PLL_S0

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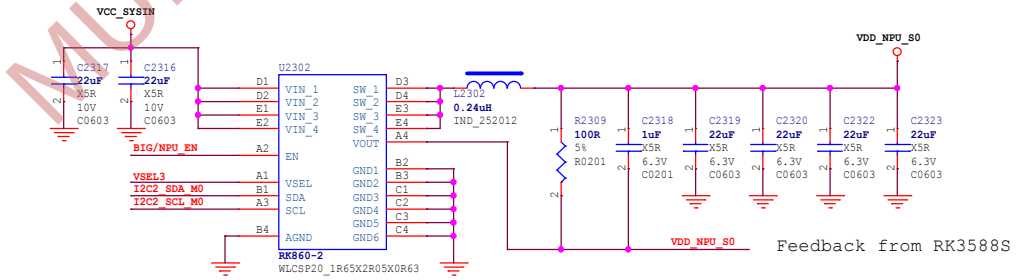
VDD_CPU_BIG0



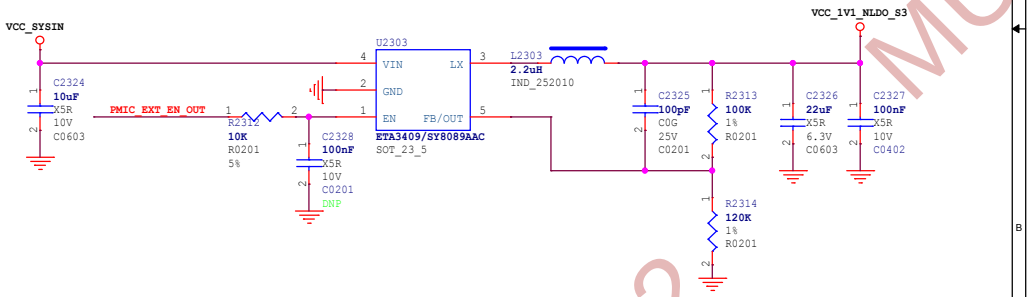
VDD_CPU_BIG1



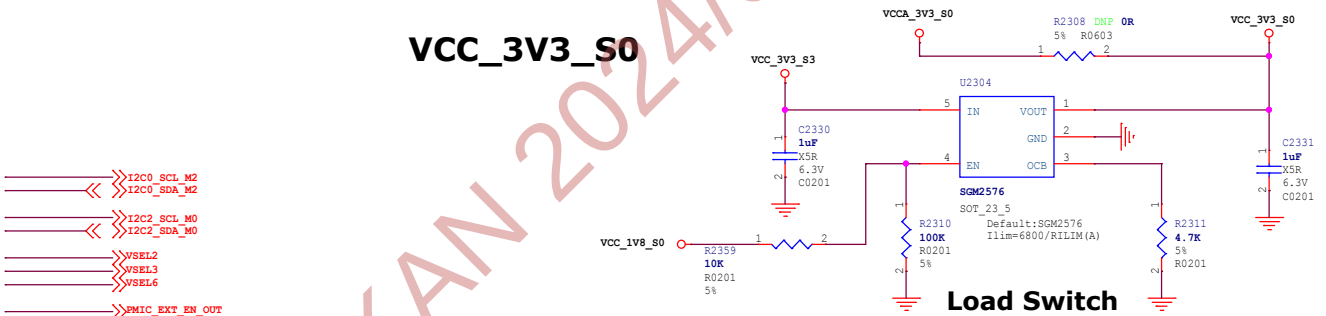
VDD_NPU



VCC_1V1_NLDO

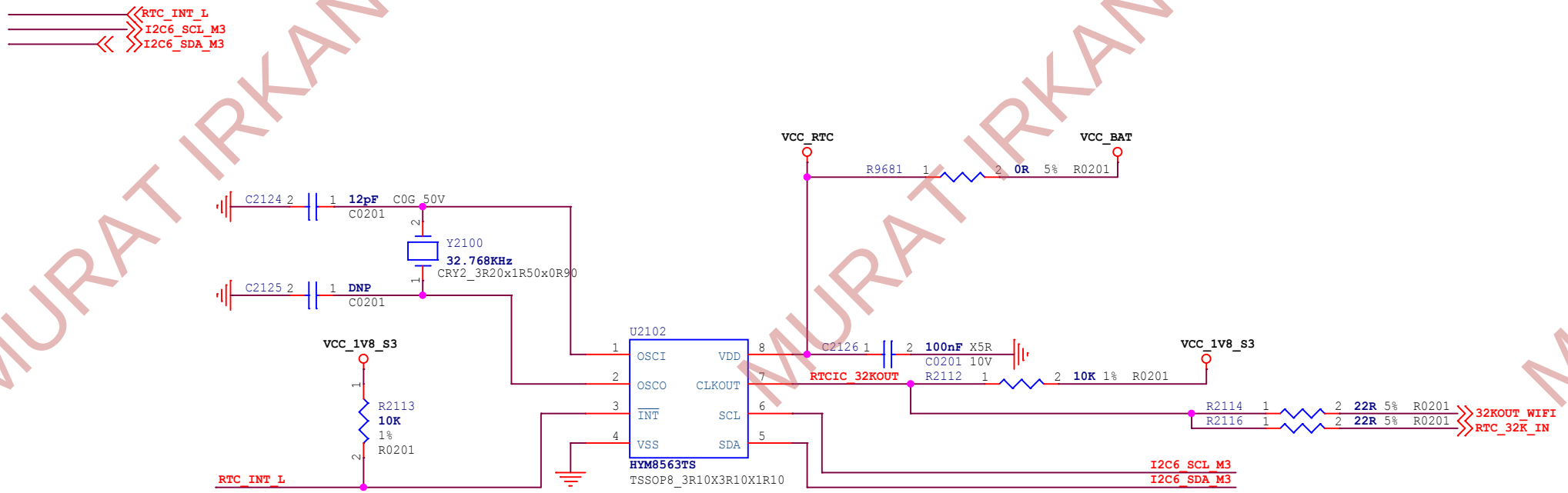


VCC_3V3_S0



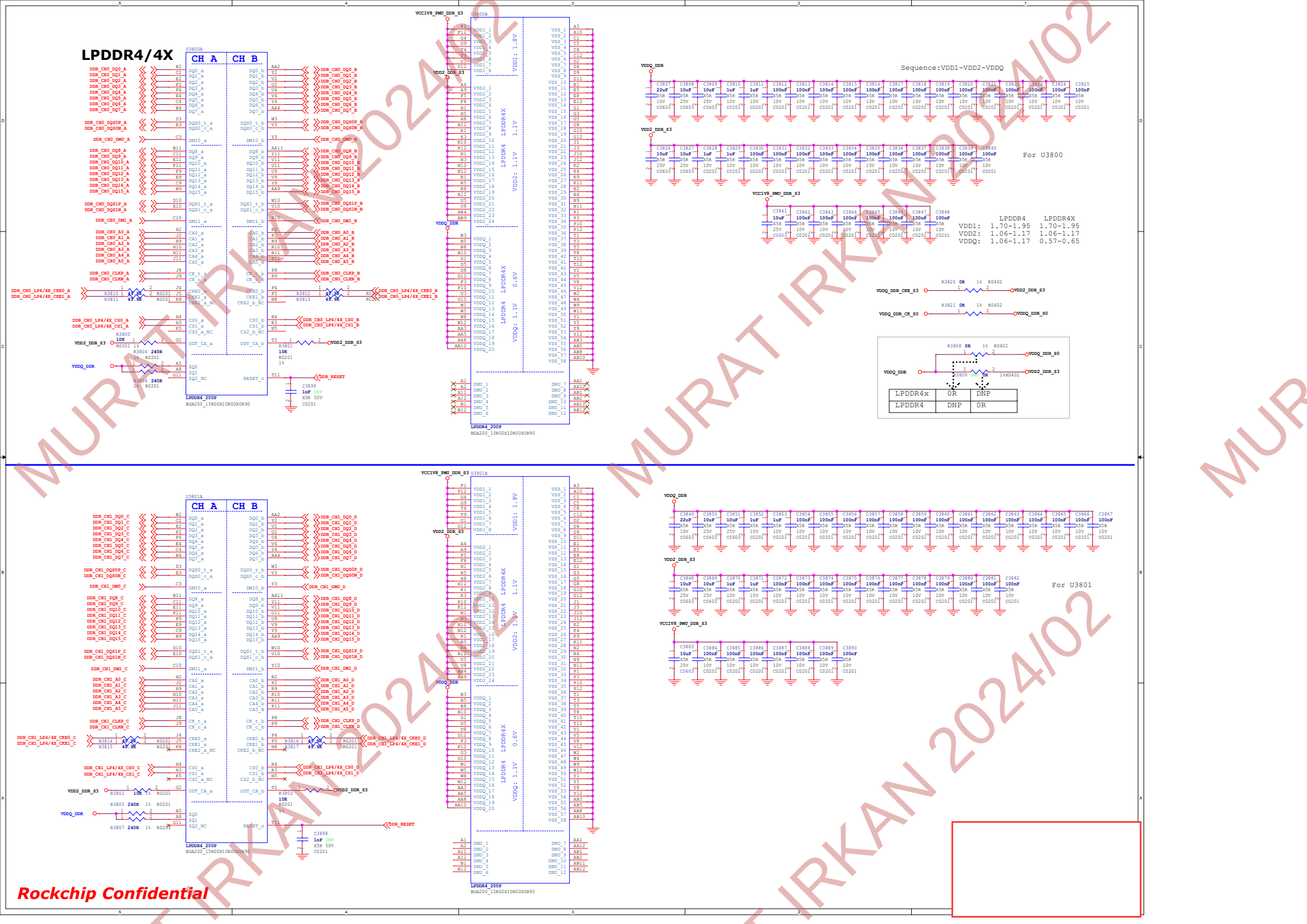
MURAT IRKAN 2024/02

RTC IC

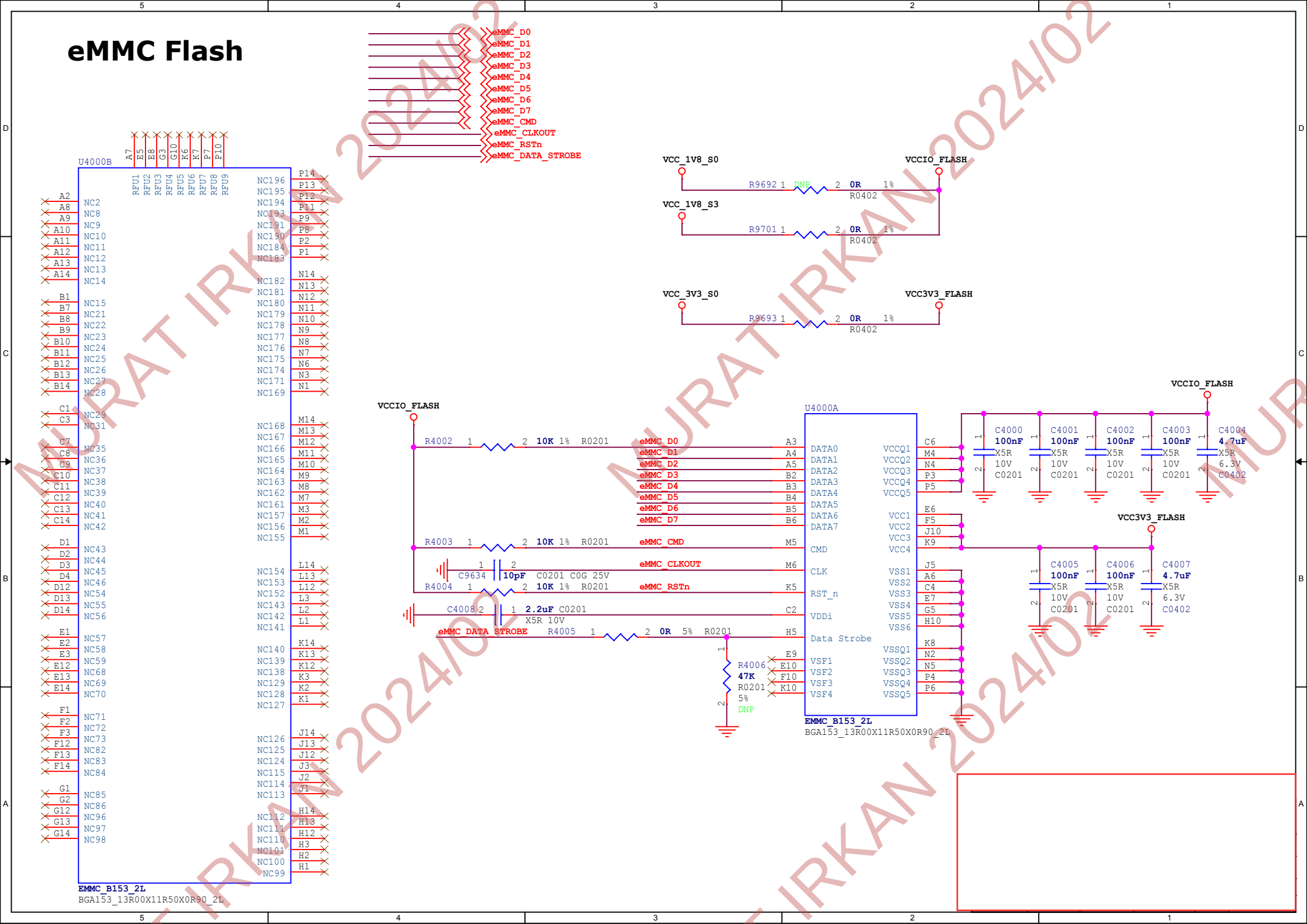


Address:Read A3H,Write A2H

MURAT IRKAN 2024/02


LPDDR4/4X
CH A
CH B
U3800A
U3800B
LPDDR4_200P
BGA200_15R00X10R00X0R90
VCC1V8_PMU_DDR_S3
VDD2_DDR_S3
VDDQ_DDR
VDD1: 1.8V
VDD2: 1.1V
VDDQ: 1.1V
LPDDR4X
LPDDR4
0.6V
Sequence:VDD1-VDD2-VDDQ
For U3800
LPDDR4 LPDDR4X
VDD1: 1.70-1.95 1.70-1.95
VDD2: 1.06-1.17 1.06-1.17
VDDQ: 1.06-1.17 0.57-0.65
VDDQ_DDR_CKE_S3
VDDQ_DDR_CK_S0
VDDQ_DDR_S0
VDDQ_DDR_S3
DDR_RESET
LPDDR4x 0R DNP
LPDDR4 DNP 0R
U3801A
U3801B
For U3801
Rockchip Confidential

eMMC Flash

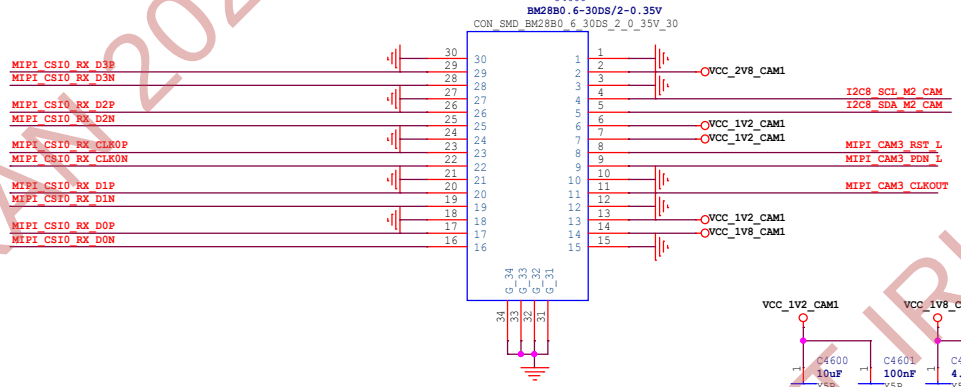


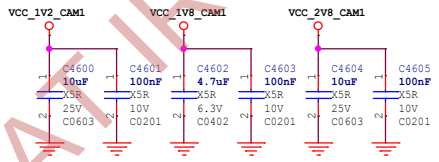
MURAT IRKAN 2024/02

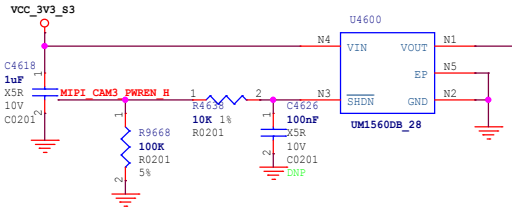
MIPI-CSI0_RX

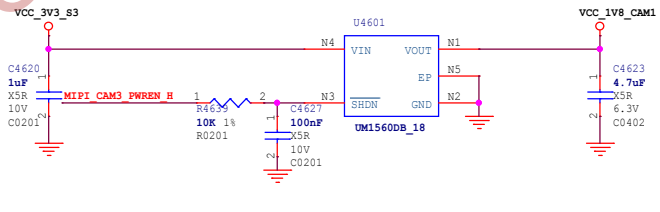
MIPI_CSI0_RX_D0P
MIPI_CSI0_RX_D0N
MIPI_CSI0_RX_D1P
MIPI_CSI0_RX_D1N
MIPI_CSI0_RX_D2P
MIPI_CSI0_RX_D2N
MIPI_CSI0_RX_D3P
MIPI_CSI0_RX_D3N
MIPI_CSI0_RX_CLK0P
MIPI_CSI0_RX_CLK0N
MIPI_CAM3_CLKOUT
MIPI_CAM3_RST_L
MIPI_CAM3_PWREN_H
MIPI_CAM3_PDN_L
I2C8_SCL_M2_CAM
I2C8_SDA_M2_CAM

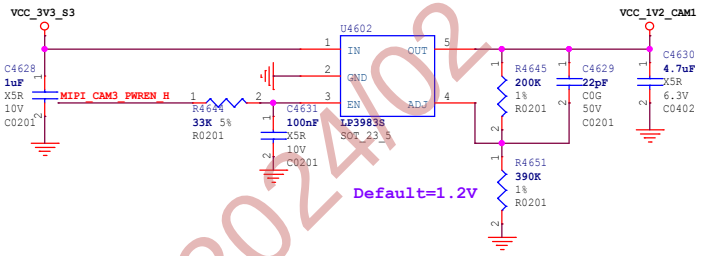
前摄： OV16A10-GA5A-Z







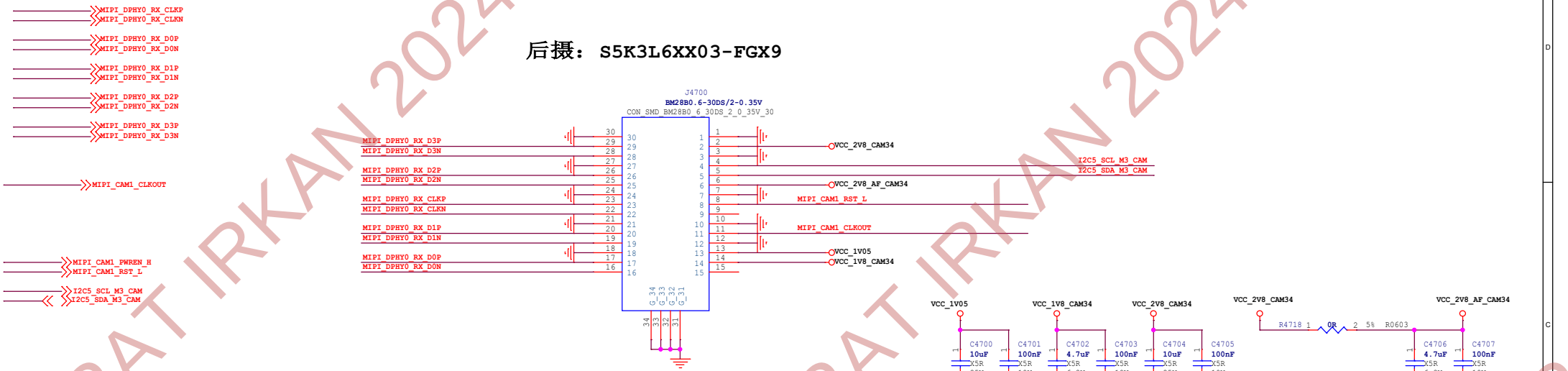


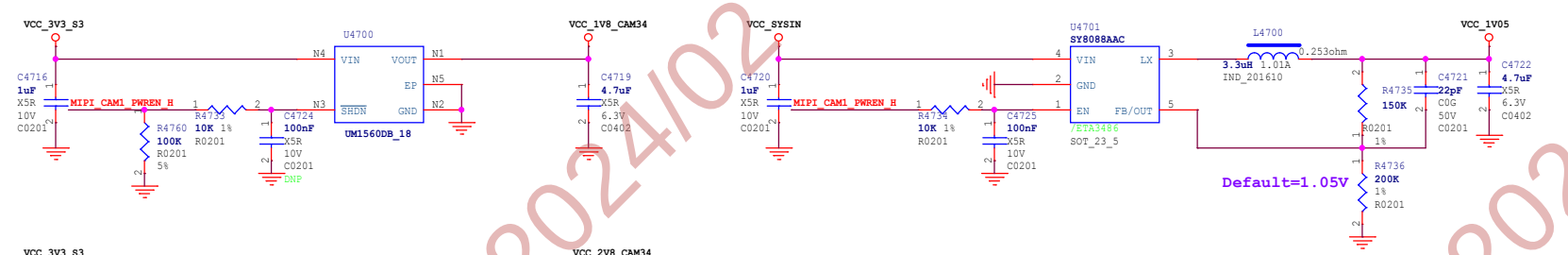


MURAT IRKAN 2024/02

VI-Camera DPHY_RX

后摄： S5K3L6XX03-FGX9

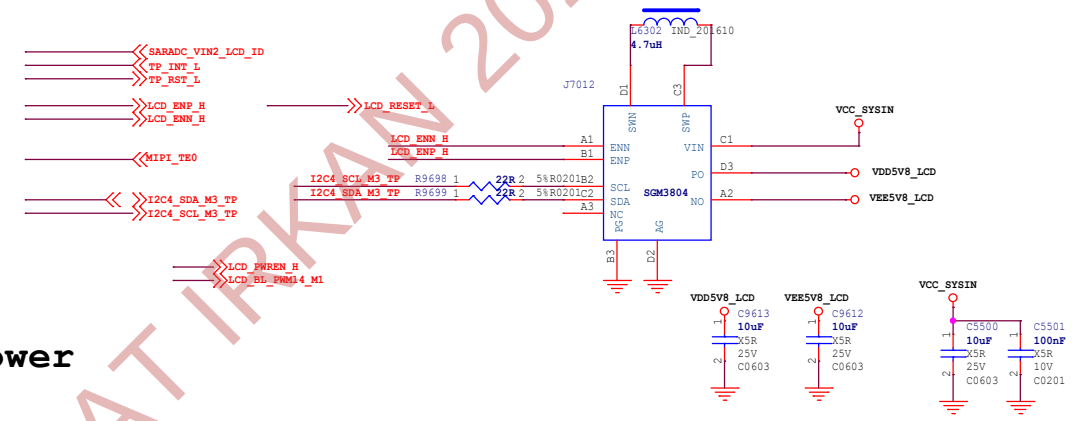


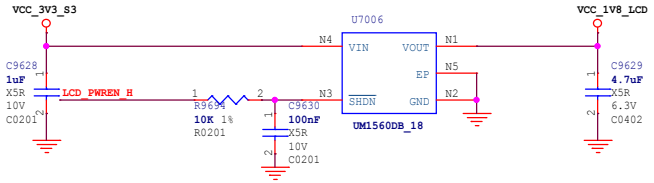


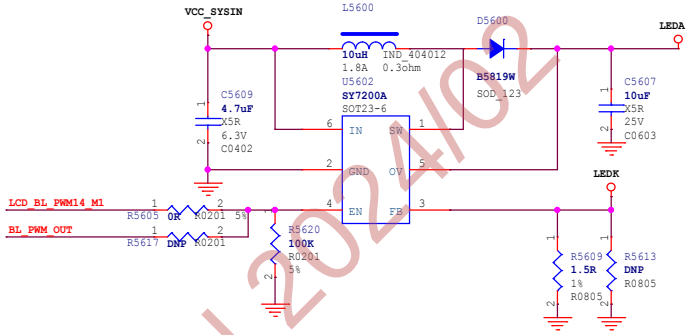
MURAT IRKAN 2024/02

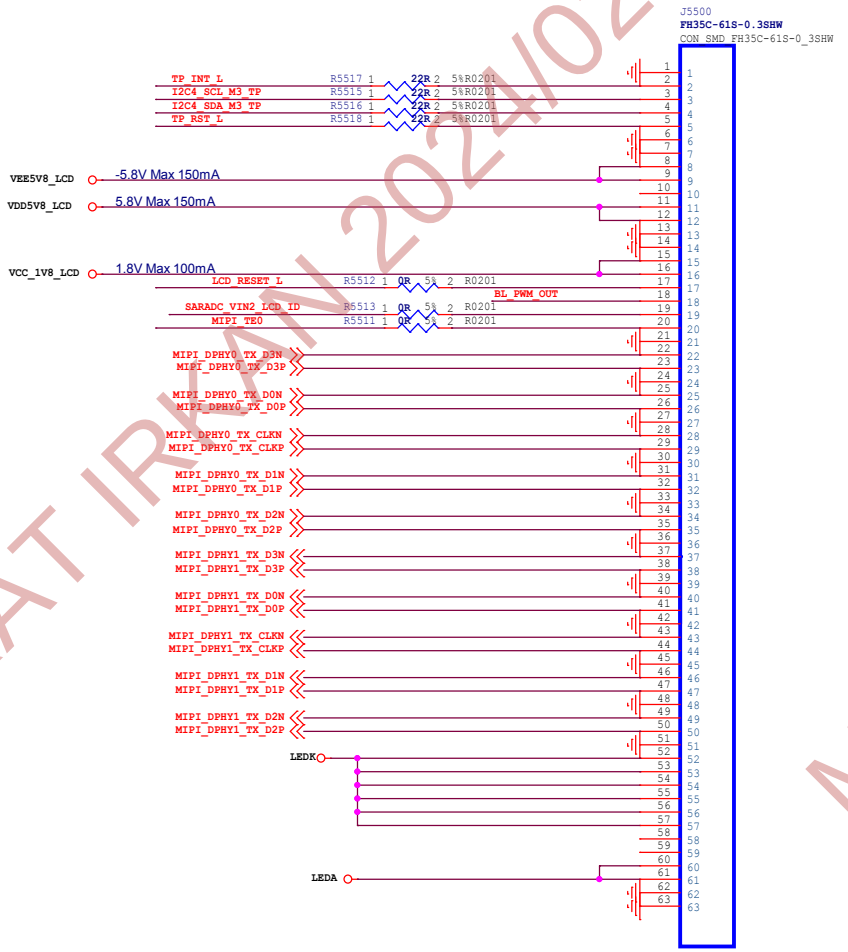
MIPI DPHY TX

Power



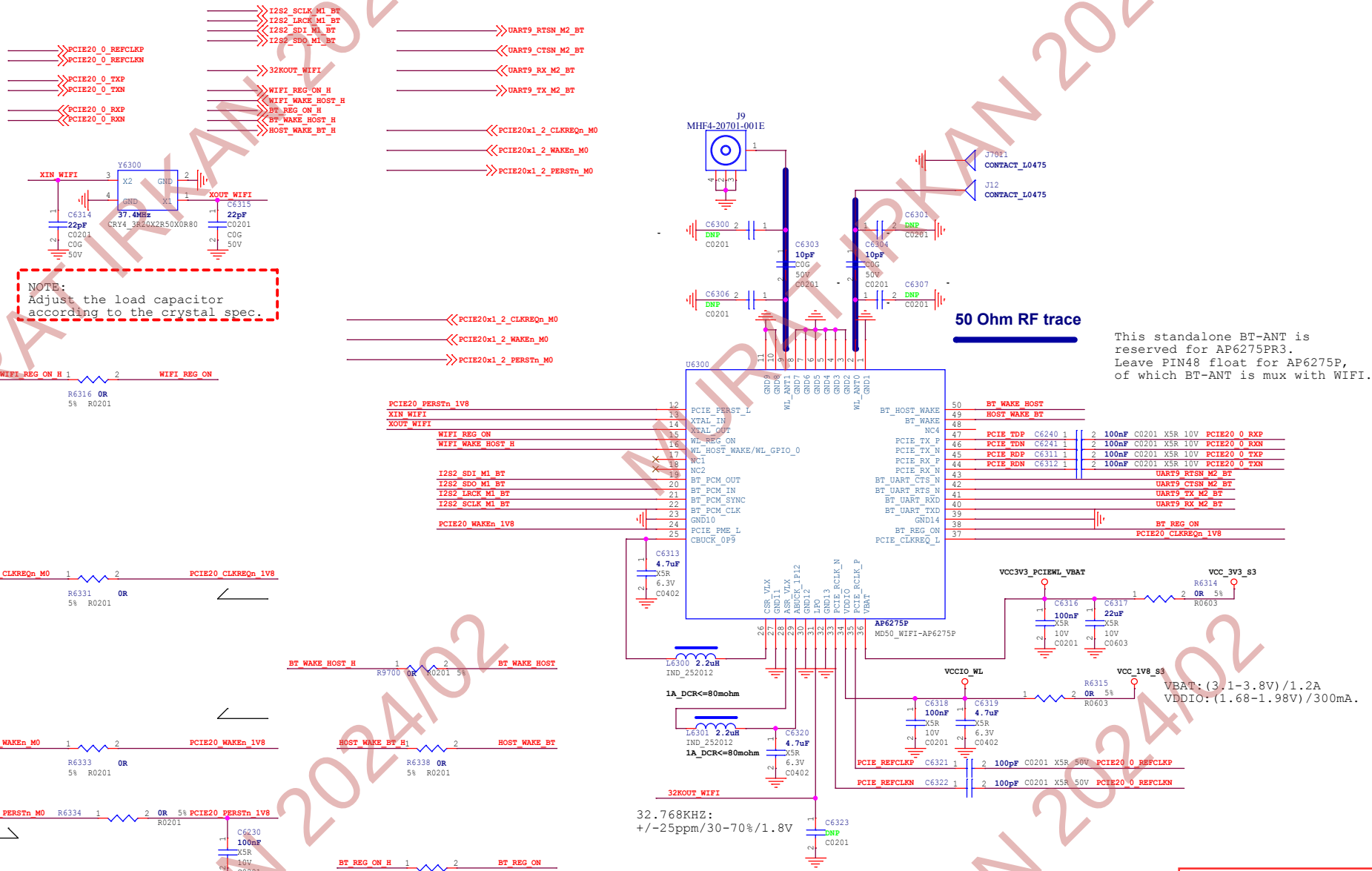




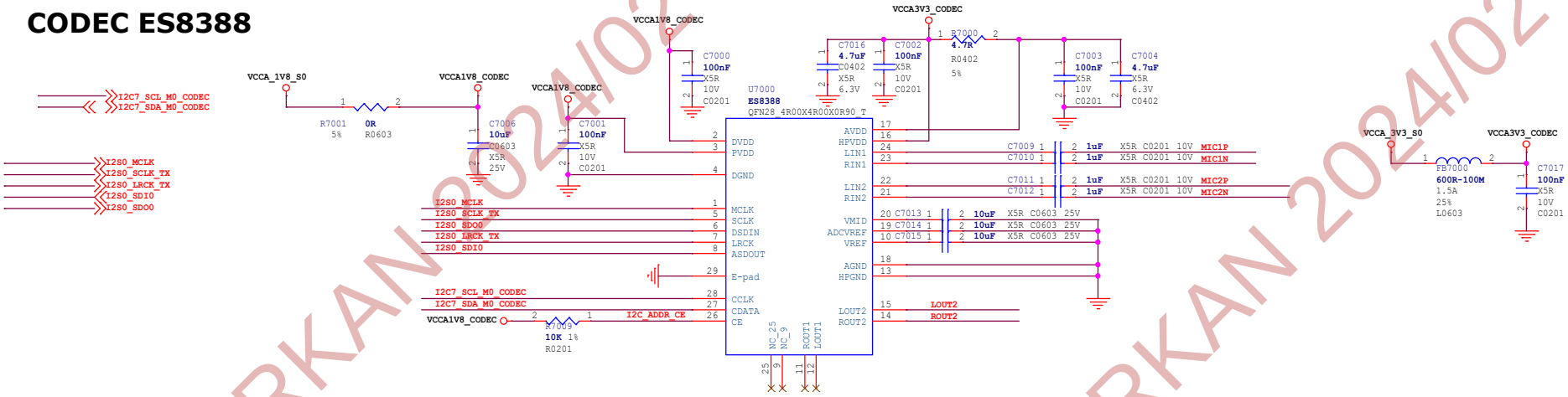


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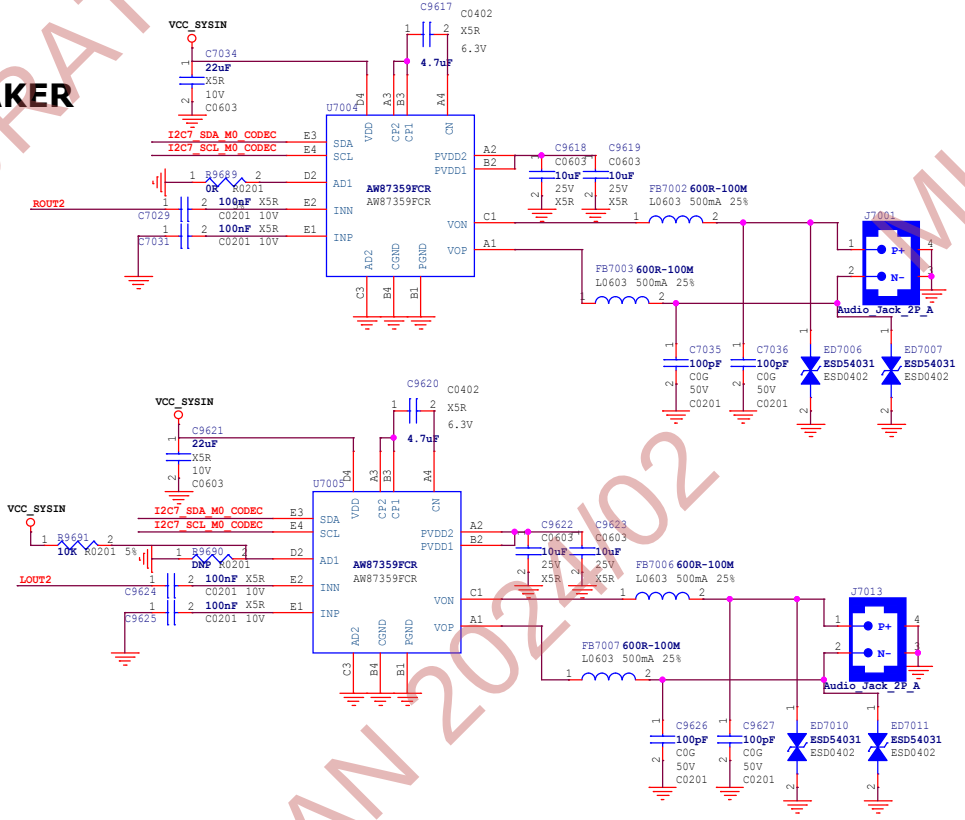
PCIe WIFI6/BT Module-2T2R



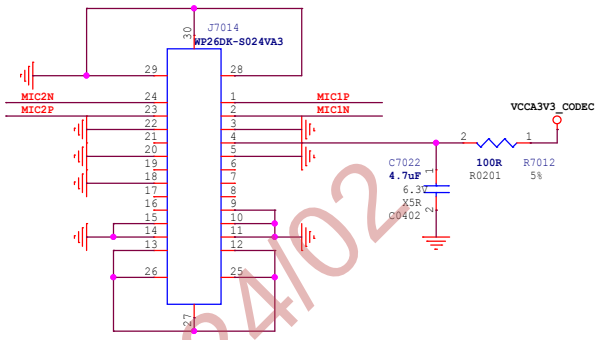
CODEC ES8388



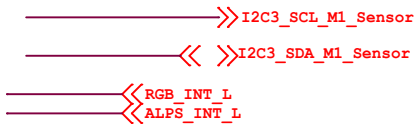
SPEAKER



Analog MIC

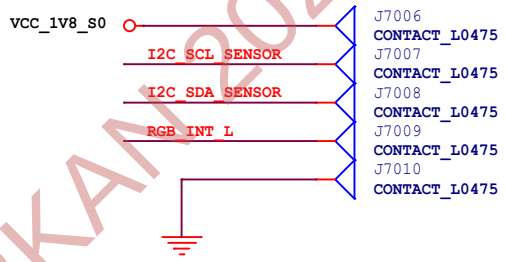


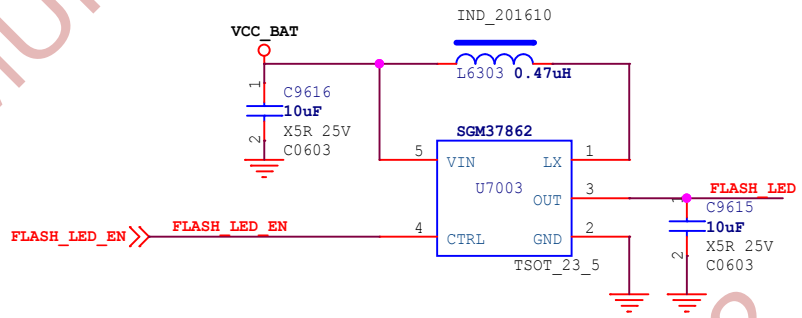
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Sensor





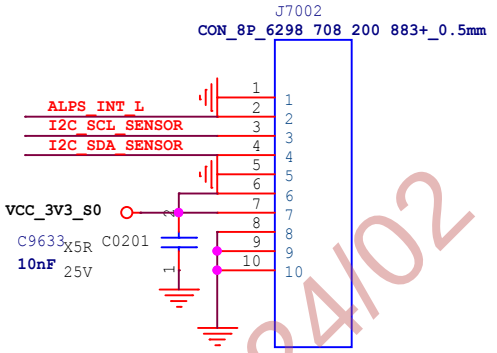


MAX:1.5A



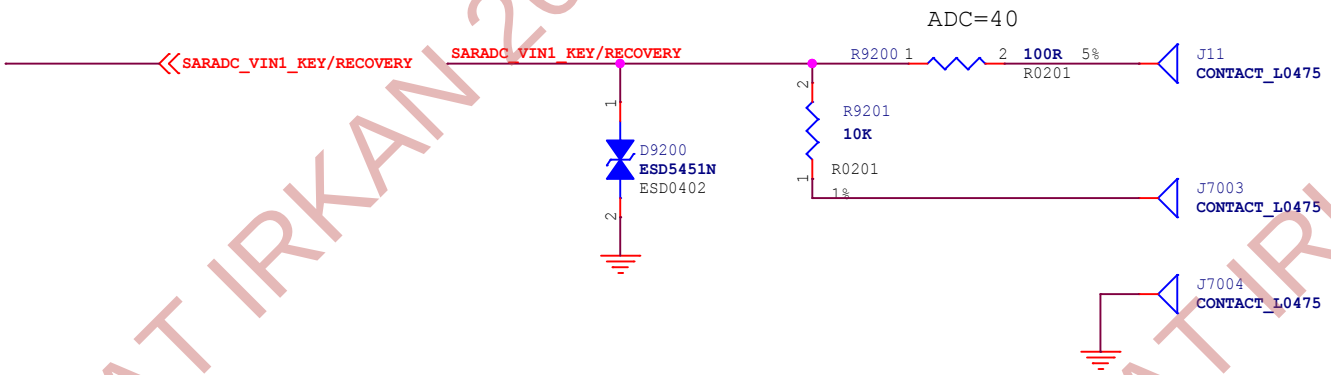
Flashlight

PLS+ALS



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KEY Array

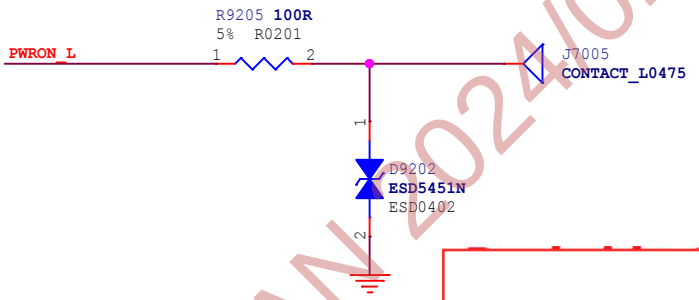


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RESET_L
PWRON_L

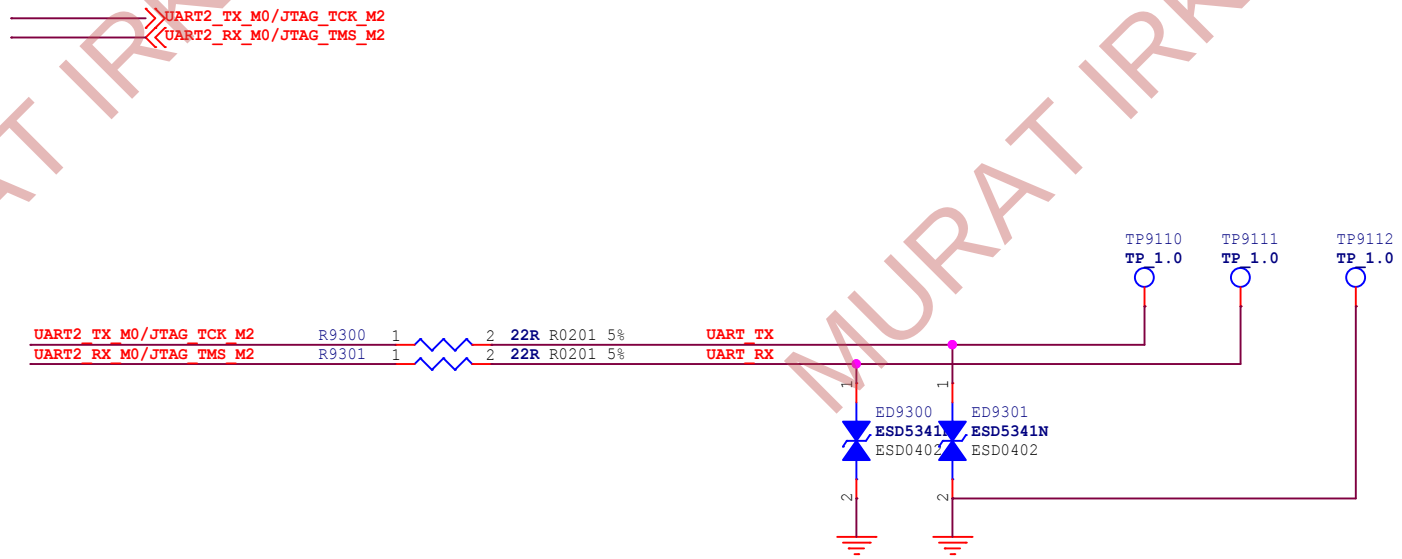
Reset_Key

PWR_Key



UART Debug

JTAG Debug



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